

目次

0.1	ソーシャルメディアと心の健康	2
0.2	人工知能と創造性	6
0.3	ファストファッションの問題点	9
0.4	昆虫の擬態	12
0.5	芸術の定義と役割	15
0.6	長文読解問題 6 大統領選：ジャクソン派とハミルトン派	18
0.7	ハリケーンと台風	21
0.8	ブラックフライデーの起源と影響	25
0.9	宇宙開発競争	29
0.10	小説『宇宙戦争』と社会的影響	33
0.11	文化相対主義と自民族中心主義	37
0.12	ナスカの地上絵	41
0.13	インカ帝国	45
0.14	大航海時代と対数の歴史	49
0.15	超伝導とリニアモーターカー	53
0.16	日米の高校生活の対比	57
0.17	大学の教育システム	61
0.18	ローマ帝国の衰退	65
0.19	日本の高度経済成長	69
0.20	日本の四大公害病	73
0.21	AI と倫理	77
0.22	流行と集団心理	81
0.23	日本のアニメの海外影響	85
0.24	おまけと社会問題	89
0.25	クリントンとブッシュの移行期	93
0.26	世界の原発事故	97
0.27	ベトナム戦争とペンタゴン・ペーパーズ	101
0.28	パディントンという名のクマ	105
0.29	彗星の謎	109
0.30	量子力学と半導体	113
0.31	相対性理論と GPS	117
0.32	東京タワーの歴史	121
0.33	ハーバー・ボッシュ法	125
0.34	共生、片利共生、寄生	129
0.35	地震とその対策	133

0.1 ソーシャルメディアと心の健康

次の英文を読み、後の設問に答えなさい。

The Double-Edged Sword: Social Media and Teen Mental Health

〈In today's interconnected world,〉 social media platforms have become a central part of life for many teenagers. They offer exciting ways to connect with friends, share experiences, and access information. (However, the constant exposure to a digital world also presents unique challenges to their mental well-being. Understanding both the positive and negative impacts is crucial for navigating this complex landscape.

On the one hand, social media can provide significant benefits. For teens who feel isolated, it can be a gateway to finding supportive communities and making friends who share similar interests or experiences. These online networks can offer a sense of belonging and reduce feelings of loneliness. Furthermore, these platforms can be powerful tools for social activism and raising awareness about important issues, empowering young people to make a positive impact on the world.

On the other hand, the negative consequences can be serious. One of the most well-known risks is cyberbullying, which can have lasting emotional effects. Another significant pressure comes from social comparison. Teens may constantly see curated, idealized versions of their peers' lives, leading to feelings of inadequacy and low self-esteem. This can also fuel a "Fear of Missing Out" (FOMO), an anxiety that they are being excluded from rewarding experiences that others are having. Finally, the use of social media, especially late at night, can disrupt sleep patterns, which is vital for mental and physical health.

In conclusion, social media is a double-edged sword. While it offers valuable opportunities for connection and empowerment, it also carries risks like cyberbullying, social comparison, and sleep disruption. The key for teenagers is not necessarily to avoid social media altogether, but to learn how to use it in a mindful and balanced way, prioritizing their real-life well-being.

設問

問 1. What is the main idea of the passage?

- (a) The various ways teenagers can become famous on social media.
- (b) The positive and negative effects of social media on teenagers' mental health.
- (c) Why teenagers should completely stop using all social media platforms.
- (d) The technical skills required to use social media effectively.

問 2. According to the passage, which of the following is a potential benefit of social media for teenagers?

- (a) It always improves their sleep quality.
- (b) It prevents them from ever feeling lonely.
- (c) It allows them to find others with similar interests and feel a sense of belonging.
- (d) It guarantees that they will be safe from all forms of bullying.

問 3. What does the passage suggest is a major cause of 'social comparison'?

- (a) Seeing the idealized and curated lives of others online.
- (b) Engaging in social activism with friends.
- (c) Learning new information about important issues.
- (d) Spending too much time with friends in real life.

問 4. The author uses the phrase 'a double-edged sword' to describe social media. What does this mean?

- (a) It is something that is brand new and exciting.
- (b) It is something that is very dangerous and should be avoided.
- (c) It is something that is difficult to understand.
- (d) It is something that has both positive and negative consequences.

問 5. In the third paragraph, what is the closest meaning of the underlined word "curated"?

- (a) carefully chosen and organized
- (b) completely natural and unplanned
- (c) old and no longer useful
- (d) secret and hidden from others

curate ≡ 加える

解答と解説

問 1

正解

(b) The positive and negative effects of social media on teenagers' mental health.

解説

この問題は、文章全体の **main idea** (主題) を問うています。主題は、筆者が最も伝えたい中心的な考えのことです。正解の根拠は、本文が第 2 段落でプラスの影響 (benefits)、第 3 段落でマイナスの影響 (negative consequences) を論じ、両側面からテーマを掘り下げている点にあります。

問 2

正解

(c) It allows them to find others with similar interests and feel a sense of belonging.

解説

この問題は、ソーシャルメディアの **potential benefit** (潜在的な利点) を問うています。第 2 段落に「...finding supportive communities and making friends who share similar interests... These online networks can offer a sense of belonging...」と明記されており、(c) の内容と完全に一致します。

問 3

正解

(a) Seeing the idealized and curated lives of others online.

解説

この問題は、「**social comparison** ()」の **major cause** (主な原因) を問うています。第 3 段落に「*Teens may constantly see curated, idealized versions of their peers' lives, leading to feelings of inadequacy...*」とあり、オンライン上で理想化され、作り上げられた他者の生活を見ることが原因だと述べられています。(a) が正解です。

問 4

正解

(d) It is something that has both positive and negative consequences.

解説

この問題は、イディオム「**a double-edged sword** (諸刃の剣)」の意味を問うています。一つのものが良い面と悪い面の両方を併せ持つことを意味する比喩表現です。本文の文脈（プラス面とマイナス面を論じた後の結論）から、(d) が最も適切です。

問 5

正解

(a) carefully chosen and organized

解説

この問題は、第 3 段落の下線部の単語 "curated" の意味を問うています。**curate** ((動詞) キュレートする) は、自分の投稿する写真や文章を、自分の生活がより良く見えるように「意図的に選び抜き、編集・加工して見せる」というニュアンスで使われます。本文の文脈に最も合うのは「注意深く選ばれ、構成された」という意味の (a) です。

0.2 人工知能と創造性

次の英文を読み、後の設問に答えなさい。

Artificial Intelligence: A New Partner in Creativity?

AIについて説明
 When we think of Artificial Intelligence (AI), we often imagine robots performing complex calculations or analyzing huge amounts of data. However, AI is now stepping into a field once thought to be uniquely human: creativity. From composing music to generating stunning works of art, AI is challenging our ideas about where creativity comes from. 気絶するほど美しい informal / 口語的

This new form of creation is made possible by advanced algorithms known as generative models. These models are trained on vast datasets of existing images, texts, or sounds. By learning the patterns and styles from this data, the AI can then produce entirely new content. For example, an artist could use AI to create a painting in the style of a famous historical painter, or a musician could use it to compose a melody for a new song. stun 気絶する

問題提起
 Of course, this raises important questions. Can a piece of art created by an algorithm truly be considered creative? Who is the author: the AI, the programmer who designed it, or the user who gave the instructions? These are complex ethical and philosophical issues that we are just beginning to explore. There are also concerns about AI potentially replacing human artists and writers in the future.

展望・答え
 Despite these concerns, many experts see AI not as a replacement for human creativity, but as a powerful new tool. Like the invention of the camera, AI can augment our own creative abilities and open up new possibilities for expression. The future of art may not be human versus machine, but rather a collaboration between them, pushing the boundaries of what we can imagine and create. The potential for this partnership seems limitless.

設問

問 6. What is the main topic of this passage?

- (a) The history of how robots were invented.
- (b) The role of AI in creative fields and the questions it raises.
- (c) The technical details of how AI algorithms are programmed.
- (d) The reasons why human artists are better than AI.

問 7. According to the second paragraph, how does AI create new content?

- (a) By copying one existing piece of art exactly.
- (b) By learning patterns from large amounts of data.

- (c) By asking human artists for new ideas.
- (d) By using its own emotions and feelings.

問 8. In the final paragraph, what is the closest meaning of the underlined word “potential”?

- (a) problem
- (b) possibility
- (c) history
- (d) reason

問 9. What is the author’s view on the future of AI and creativity?

- (a) AI will completely replace all human artists and musicians.
- (b) Humans and AI will likely work together to create new forms of art.
- (c) The use of AI in art should be banned to protect human jobs.
- (d) AI is not advanced enough to be useful in any creative field.

解答と解説

問 6

正解

(b) The role of AI in creative fields and the questions it raises.

解説

文章は、AI が音楽やアートといった創造分野でどのような役割を果たすか（第 1・2 段落）、そしてそれが「創造性とは何か？」といった倫理的・哲学的な問題を引き起こしていること（第 3 段落）について述べています。全体を最も的確に要約しているのは (b) です。

問 7

正解

(b) By learning patterns from large amounts of data.

解説

第 2 段落に、「これらのモデルは広大なデータセットで訓練される。このデータからパターンとスタイルを学習することによって、AI は全く新しいコンテンツを生み出すことができる」と明確に書かれています。

問 8

正解

(b) possibility

解説

potential は「将来的に～になりうる可能性」や「潜在的な能力」を意味します。最終段落では、人間と AI の協働が持つ「無限の可能性」について語られており、possibility が最も近い意味です。

問 9

正解

(b) Humans and AI will likely work together to create new forms of art.

解説

最終段落で、筆者は「芸術の未来は、人間対機械ではなく、むしろ両者の協力関係（collaboration）かもしれない」と述べています。AI は人間の代替物ではなく、新しいツールとして協働していくという見解です。

0.3 ファストファッションの問題点

次の英文を読み、後の設問に答えなさい。

Fast Fashion: Trendy but Troubling

Fast fashion refers to the business model of producing inexpensive clothing rapidly in response to the **latest trends**. For many young consumers, the appeal is obvious: they can buy stylish new clothes for very low prices. This allows them to update their **wardrobes** frequently without spending a lot of money.

However, this convenience **comes with** a significant hidden cost **to our planet and the people** who make the clothes.

The environmental impact of fast fashion is **immense**. The industry consumes an enormous amount of water to grow cotton and uses toxic chemical dyes that pollute rivers. Because the clothes are so cheap, they are often seen as **disposable**. This results in millions of tons of textile waste ending up in landfills every year, where they can take hundreds of years to **decompose**.

Furthermore, there is a serious human cost. To keep prices low, many fast fashion brands produce their garments in developing countries where labor is cheap. **Garment workers**, who are mostly women, often face **low wages, long hours, and unsafe working conditions**. The constant pressure to produce clothes quickly means that safety standards are sometimes ignored.

So, what can be done? As consumers, we have the power to make a change. By choosing to buy fewer items of higher quality, supporting sustainable brands, and shopping secondhand, we can reduce our impact. Being a conscious consumer means thinking about where our clothes come from and aiming for a fashion industry that is fair to both people and the planet.

設問

問 10. What is the main purpose of this passage?

- (a) To explain how to become a famous fashion designer.
- (b) To advertise the benefits of buying inexpensive clothing.
- (c) To raise awareness about the negative consequences of fast fashion.
- (d) To compare fashion trends in different countries.

問 11. According to the passage, which of the following is an environmental problem caused by fast fashion?

- (a) It helps to clean rivers by removing toxic chemicals.
- (b) It leads to a large amount of textile waste in landfills.
- (c) It uses very little water to produce clothing.

- (d) It encourages the recycling of all clothing materials.

問 12. In the final paragraph, what is the best definition for the underlined word “sustainable”?

- (a) very fashionable and popular
- ☒ (b) able to be maintained without harming the environment
- (c) extremely cheap and easy to produce
- (d) traditional and old-fashioned

問 13. What does the author suggest consumers should do?

- (a) Buy new clothes every week to follow the latest trends.
- (b) Only purchase clothes from the most expensive luxury brands.
- ☒ (c) Think more carefully about their shopping choices and their impact.
- (d) Avoid buying secondhand clothes as they are not clean.

解答と解説

問 10

正解

(c) To raise awareness about the negative consequences of fast fashion.

解説

この文章は、ファストファッションが環境や労働者に与える「隠れたコスト」に焦点を当て、その問題点を読者に知らせることを目的としています。

問 11

正解

(b) It leads to a large amount of textile waste in landfills.

解説

第2段落に、「何百万トンもの繊維ゴミが毎年、最終的に埋め立て地に送られることになる」と、ファストファッションによる環境問題として具体的に挙げられています。

問 12

正解

(b) able to be maintained without harming the environment

解説

sustainable は、特に環境問題の文脈で、「環境を破壊することなく、将来にわたって維持できる」という意味で使われます。

問 13

正解

(c) Think more carefully about their shopping choices and their impact.

解説

最終段落で、「意識的な消費者であることは、自分の服がどこから来るのかについて考えること」だと述べ、質の高いものを買うことや古着を選ぶことなど、より慎重な選択を促しています。

0.4 昆虫の擬態

次の英文を読み、後の設問に答えなさい。

The Art of Deception: Mimicry in the Insect World

In the natural world, survival is a constant challenge. To avoid becoming a meal for a hungry predator, insects have developed a remarkable range of defensive strategies. Among the most fascinating of these is mimicry, the ability of one species to imitate another, or even a non-living object, to gain an advantage. This strategy is a sophisticated form of deception that has evolved over millions of years.

One common type of mimicry is called Batesian mimicry. This occurs when a harmless species imitates a dangerous or unpalatable one. For example, some species of hoverflies have evolved to look like wasps or bees. These hoverflies are defenseless, but their appearance tricks predators into thinking they are dangerous and should be avoided. The success of this mimicry depends entirely on the predator's prior negative experience with the real, dangerous insect.

Another form is Müllerian mimicry. This is a cooperative strategy where two or more species that are both dangerous or distasteful to predators have evolved to look similar. The best-known example is the coloration of various poisonous butterflies. By sharing the same warning signs, predators only need to learn to avoid one type of insect to know that all similarly colored insects are inedible. This reinforces the lesson for the predator, benefiting all species involved.

In addition to imitating other insects, some species use camouflage to blend into their surroundings. The stick insect, for instance, looks exactly like a twig, making it nearly impossible for predators to spot. This ability allows it to remain motionless during the day and move only when it is safely hidden. Whether through imitation or camouflage, these intricate strategies highlight the incredible diversity of life and the endless creativity of evolution. The study of these behaviors is a key part of entomology, the scientific study of insects.

設問

問 14. What is the main idea of the passage?

- (a) The history of how insects have evolved.
- (b) The different types of camouflage used by insects.
- (c) The various types of mimicry and camouflage insects use to survive.
- (d) The reasons why hoverflies are dangerous to predators.

問 15. According to the passage, what is a key difference between Batesian mimicry and Müllerian

mimicry?

- (a) ~~Only~~ Batesian mimicry involves imitating a dangerous species.
- (b) In Batesian mimicry, the imitating species is harmless, while in Müllerian, both are dangerous.
- (c) Batesian mimicry is a cooperative strategy, while Müllerian is not.
- (d) Müllerian mimicry is more effective for avoiding predators.

問 16. How does a stick insect avoid being eaten by predators?

- (a) It makes a loud noise to scare them away.
- (b) It imitates a dangerous butterfly.
- (c) It uses its camouflage to look like a twig.
- (d) It flies very fast to escape.

問 17. Based on the final paragraph, which of the following best defines the term “entomology”?

- (a) The study of animal evolution.
- (b) The scientific study of insects.
- (c) The art of deceiving predators.
- (d) The research of plant life.

解答と解説

問 14

正解

(c) The various types of mimicry and camouflage insects use to survive.

解説

この文章は、昆虫が捕食者から身を守るために使う「擬態 (Batesian mimicry、Müllerian mimicry)」と「保護色 (camouflage)」という、複数の戦略について説明しています。文章全体を最もよく要約しているのは (c) です。

問 15

正解

(b) In Batesian mimicry, the imitating species is harmless, while in Müllerian, both are dangerous.

解説

第2段落に Batesian mimicry は「無害な種が危険な種を模倣する」とあり、第3段落に Müllerian mimicry は「両方とも危険な、または不味い2種以上の種が互いに似た外見に進化する」と書かれています。(b) が両者の違いを最も正確に説明しています。

問 16

正解

(c) It uses its camouflage to look like a twig.

解説

最終段落に「ナナフシは、枝そっくりに見える保護色 (camouflage) を使う」と記述されています。これにより、捕食者に見つけられるのを避けることができると説明されています。

問 17

正解

(b) The scientific study of insects.

解説

最終段落の最後の文に、「entomology (昆虫学) とは、昆虫の科学的研究である」と定義が明記されています。(b) がこの定義と一致します。

0.5 芸術の定義と役割

次の英文を読み、後の設問に答えなさい。

What is Art? Beyond the Canvas and Clay

Art is a concept that has fascinated humanity for centuries. While we often associate it with masterpieces in museums, such as a painting by Leonardo da Vinci or a sculpture by Michelangelo, its true definition is much broader. Art is fundamentally a form of human expression. It can be a way to communicate thoughts, emotions, or ideas that are difficult to put into words. This is why a simple folk song, a beautifully decorated ceramic bowl, or even a piece of well-designed furniture can be considered art.

文化
Historically, art has served many different purposes. In ancient civilizations, it was often functional, used for religious rituals, storytelling, and documenting history. The cave paintings of Lascaux, for example, are not just beautiful images, but also provide valuable insights into the daily lives and beliefs of early humans. In later periods, art became a symbol of power and wealth, commissioned by kings and the church to demonstrate their authority and piety.

In the modern world, the role of art has expanded even further. It now serves as a powerful tool for social commentary and change. Artists use their work to challenge social norms, provoke thought, and raise awareness about important issues. A powerful photograph of a historical event, a film that explores a complex social problem, or a street mural that inspires hope can all be forms of art that shape public opinion. The beauty of art is its ability to hold a mirror up to society, reflecting both its triumphs and its flaws.

In conclusion, art is not limited to just famous works in galleries. It is a universal language that has evolved from a functional tool to a complex form of self-expression and social critique. It reminds us of our shared humanity and pushes us to see the world from different perspectives. As long as humans have stories to tell and emotions to share, there will always be a place for art in our lives. The essence of art is its enduring power to connect us all.

設問

問 18. What is the main idea of the passage?

- (a) The most famous artists in history are from Europe.
- (b) Art is a complex concept that has evolved in its definition and purpose over time.
- (c) Ancient art is more important than modern art.
- (d) Art is only found in museums and galleries.

問 19. According to the second paragraph, what was a purpose of art in ancient times?

- (a) To make a profit from selling paintings.
- (b) To inspire political revolution.
- (c) To be a tool for religious rituals and storytelling.
- (d) To be a way for artists to become famous.

問 20. What does the author suggest is a modern role of art?

- (a) To document the private lives of wealthy people.
- (b) To serve as a tool for social commentary and change.
- (c) To be used only for religious ceremonies.
- (d) To be a form of entertainment with no deeper meaning.

問 21. What is the closest meaning of the underlined word “essence” in the final paragraph?

- (a) a large collection of something
- (b) the history of something
- (c) the fundamental nature or most important quality of something
- (d) a beautiful appearance

解答と解説

問 18

正解

(b) Art is a complex concept that has evolved in its definition and purpose over time.

解説

この文章は、芸術が「博物館にある作品」といった狭い定義を超え、時代とともにその目的や役割を変化させてきたことを説明しています。古代の機能的な役割から、現代の社会的な役割まで、その変遷を論じているため、(b) が主題に最も適しています。

問 19

正解

(c) To be a tool for religious rituals and storytelling.

解説

第2段落に、「古代文明においては、芸術はしばしば機能的であり、**宗教的な儀式や物語を伝えるため**に用いられた」と明確に書かれています。

問 20

正解

(b) To serve as a tool for social commentary and change.

解説

第3段落に、「現代においては、芸術は**社会的な論評や変化のための強力なツール**として機能する」と述べられており、社会規範に異議を唱え、重要な問題への意識を高める役割があると説明されています。

問 21

正解

(c) the fundamental nature or most important quality of something

解説

‘essence’は「本質、最も重要な要素」を意味する単語です。文章の最後で、「芸術の**本質**は、私たち全員をつなぐその永続的な力だ」と結論づけているため、(c) が最も適切な定義です。

0.6 長文読解問題 6 大統領選：ジャクソン派とハミルトン派

次の英文を読み、後の設問に答えなさい。

Two Visions for America: The Legacy of Jackson and Hamilton

In the early history of the United States, two powerful but contrasting visions emerged for the nation's future. These ideas, primarily championed by Alexander Hamilton and Andrew Jackson, laid the foundations for political debates that continue to shape American politics today. While neither figure would recognize the modern political parties, their core principles and disagreements provide a framework for understanding the country's development.

Alexander Hamilton, a key figure among the Founding Fathers, was a strong believer in a powerful central government. He envisioned a nation with a thriving industrial economy, supported by a strong national bank and a high tariff to protect American industries. Hamilton's supporters, often merchants and wealthy landowners, believed that a centralized government was essential for economic stability and national strength. His policies were designed to favor a commercial elite and promote order and efficiency.

In stark contrast, Andrew Jackson, the seventh president, was a populist leader who distrusted centralized power. He saw the national bank as a tool for the wealthy to exploit the common people and advocated for the rights of the "common man," including farmers and frontier settlers. Jackson's supporters believed that government should be closer to the people and that democracy should be expanded. His vision emphasized individual liberty and a limited federal government, focusing on the needs of the agricultural sector over industrial growth. The conflict between these two philosophies was most famously seen in Jackson's destruction of the Second Bank of the United States.

The divide between these two philosophies—the ****Hamiltonian**** emphasis on centralized power and industrial development versus the ****Jacksonian**** focus on individual liberty and agrarian democracy—persists in American politics. While the issues have changed, the fundamental debate over the role of government and the balance between different economic interests remains. Understanding this historical opposition is crucial for anyone trying to make sense of American political history.

設問

問 22. What is the main purpose of this passage?

- (a) To argue that Andrew Jackson was a better leader than Alexander Hamilton.
- (b) To describe the history of the modern American political parties.

- ☒ (c) To analyze the two contrasting visions of Alexander Hamilton and Andrew Jackson.
- ☐ (d) To explain how the US economy was created.

問 23. According to the passage, what did Alexander Hamilton advocate for?

- ☐ (a) A weak central government.
- ☐ (b) Policies that would help frontier settlers.
- ☐ (c) An economy based on agriculture.
- ☒ (d) A strong national bank.

問 24. Which of the following best describes Andrew Jackson's view on government?

- ☐ (a) He believed that a strong government should control the economy.
- ☐ (b) He was a strong supporter of the national bank.
- ☒ (c) He distrusted centralized power and favored the "common man."
- ☐ (d) He wanted to protect the interests of wealthy merchants.

問 25. What does the author suggest about the ideas of Hamilton and Jackson today?

- ☐ (a) The political parties today have forgotten their ideas completely.
- ☐ (b) The specific issues are the same, but the names have changed.
- ☒ (c) The core conflict between their philosophies still influences modern politics.
- ☐ (d) Modern politics are more stable because of their disagreements.

解答と解説

問 22

正解

(c) To analyze the two contrasting visions of Alexander Hamilton and Andrew Jackson.

解説

この文章は、第2段落でハミルトンの「強力な中央政府」という構想を、第3段落でジャクソンの「中央集権を疑う」という構想をそれぞれ説明し、両者の対比を分析しています。文章全体の目的を最も的確に表しているのは(c)です。

問 23

正解

(d) A strong national bank.

解説

第2段落に、「ハミルトンは、強力な**国立銀行**と、アメリカの産業を保護するための高い関税によって支えられる、繁栄した工業経済を持つ国家を思い描いた」と書かれています。

問 24

正解

(c) He distrusted centralized power and favored the “common man.”

解説

第3段落の冒頭に、「対照的に、アンドリュー・ジャクソンは**中央集権的な権力を信用しない**大衆的な指導者だった」とあり、「農民や開拓者を含む『**一般人**』の権利を擁護した」と述べられています。

問 25

正解

(c) The core conflict between their philosophies still influences modern politics.

解説

最終段落に、「これらの二つの哲学間の対立... は、現代アメリカの政治にも**根強く残っている**」と書かれています。さらに、「問題は変わったものの、政府の役割と異なる経済的利益のバランスを巡る根本的な議論は続いている」と述べられており、(c)が正解です。

0.7 ハリケーンと台風

次の英文を読み、後の設問に答えなさい。

The Name is Different, the Storm is the Same: Understanding Tropical Cyclones

When a powerful storm with swirling winds develops over tropical oceans, it is given a name to help meteorologists track its path. While the names "hurricane" and "typhoon" are commonly used, they actually refer to the same type of weather phenomenon, known scientifically as a **tropical cyclone**. The only difference between them is where they form.

A storm is called a **hurricane** when it forms over the North Atlantic, the Northeast Pacific, or the Central Pacific Ocean. These are the powerful storms that can devastate coastal areas in North and Central America. The term is believed to have originated from the Spanish word 'huracán', which was inspired by a Caribbean god of wind.

A storm is called a **typhoon** when it develops in the Northwest Pacific Ocean, which includes regions near Japan, China, and the Philippines. This basin is known for producing the most intense and frequent tropical cyclones in the world. The word 'typhoon' has a fascinating etymology, possibly coming from the Chinese word 'tai feng', meaning "great wind."

Finally, if the same kind of storm forms in the Indian Ocean or the South Pacific, it is simply called a **tropical cyclone** or sometimes a **cyclonic storm**. Regardless of its name, a tropical cyclone needs warm ocean water to form and gain strength. As climate change leads to warmer oceans, scientists are concerned that these storms may become more powerful and frequent in the future, posing a greater threat to coastal populations around the world.

設問

問 26. What is the main difference between a hurricane and a typhoon?

- (a) Their size.
- (b) Their wind speed.
- (c) Their place of origin.
- (d) Their duration.

問 27. According to the passage, where does a storm have to form to be called a hurricane?

- (a) The Indian Ocean.
- (b) The Northwest Pacific Ocean.
- (c) The North Atlantic Ocean.

(d) The South Pacific Ocean.

問 28. What is the scientific term for both hurricanes and typhoons?

- (a) 'cyclonic storm'
- (b) 'tropical cyclone'
- (c) 'storm surge'
- (d) 'weather phenomenon'

問 29. Based on the passage, what is a necessary condition for a tropical cyclone to form and strengthen?

- (a) Very cold air temperatures.
- (b) An existing mountain range.
- (c) Warm ocean water.
- (d) A large landmass nearby.

解答と解説

問 26

正解

(c) Their place of origin.

解説

文章の冒頭で、「ハリケーンと台風は、**どこで発生するか**というただ一つの違いしかない」と明確に述べられています。つまり、分類される地域が唯一の相違点です。**重要単語の解説:**

origin 起源、発生地。物事がどこから始まるかを表す名詞です。

問 27

正解

(c) The North Atlantic Ocean.

解説

第2段落に、「北**大西洋**、北東太平洋、または中央太平洋で発生した嵐はハリケーンと呼ばれる」と記述されています。(c) がその条件に合致します。

問 28

正解

(b) ‘tropical cyclone’

解説

第1段落で、ハリケーンと台風は科学的に「**熱帯低気圧 (tropical cyclone) **」として知られる同じ種類の気象現象だと説明されています。**重要単語の解説:**

phenomenon 現象。特に、自然界や社会で観察される出来事を指す単語です。

問 29

正解

(c) Warm ocean water.

解説

最終段落に、「熱帯低気圧は、形成され、**勢力を増すために温かい海の水が必要だ**」と書かれています。これは、温かい海からエネルギーを得て発達するためです。**重要単語の解説:**

condition 条件。何かが起こるために満たされるべき要件を意味します。

strengthen 強化する、勢いを増す。

0.8 ブラックフライデーの起源と影響

次の英文を読み、後の設問に答えなさい。

From Thanksgiving to a Shopping Frenzy: The Story of Black Friday

Black Friday, the day after Thanksgiving in the United States, is now a global phenomenon synonymous with huge sales and deep discounts. While it has become a major shopping event, its name and purpose have evolved over time. The term did not originally refer to a day of massive profit for retailers; its history is more complex and less celebratory.

The earliest use of the term “Black Friday” to describe the day after Thanksgiving dates back to the 1950s in Philadelphia. It was used by the police to describe the chaos that ensued as large crowds of suburban shoppers and tourists flooded into the city for the annual Army-Navy football game. The term painted a picture of traffic jams, pedestrian congestion, and general disorder, making the day a nightmare for law enforcement. Retailers, unhappy with the negative connotation, tried to rebrand the day as “Big Friday,” but the new name never caught on.

By the 1980s, retailers successfully changed the meaning of the phrase to something more positive. They came up with a new explanation: it was the day that businesses finally moved from being “in the red” (losing money) to “in the black” (making a profit) for the year. This new narrative, combined with aggressive marketing campaigns, transformed Black Friday into the biggest shopping day of the year. This shift marked a fundamental change in its perception, turning a term of chaos into a symbol of economic success.

Today, Black Friday has spread far beyond the US and has become a global event, now often starting weeks before Thanksgiving. It is now followed by “Cyber Monday,” a similar event for online sales. This global expansion, however, also raises questions about consumerism and its environmental impact. The constant pressure to buy new goods contributes to overconsumption, which has significant consequences for both the planet and people.

設問

問 30. What was the original meaning of the term “Black Friday”?

- (a) A day when retailers made a lot of money.
- (b) A day of chaos and congestion for police.
- (c) The day that the American economy failed.
- (d) A day for a national football holiday.

問 31. According to the passage, who first used the term “Black Friday” in the 1950s?

- (a) Retailers in Philadelphia.
- (b) The police in Philadelphia.
- (c) The military.
- (d) Journalists.

問 32. How did retailers change the meaning of “Black Friday” in the 1980s?

- (a) By associating it with the start of the holiday season.
- (b) By linking it to businesses making a profit.
- (c) By using a new name, “Big Friday.”
- (d) By celebrating it as a day of community.

問 33. The author suggests that the modern Black Friday has a negative side. What is it?

- (a) It causes traffic jams.
- (b) It encourages consumerism and overconsumption.
- (c) It is not a major shopping day anymore.
- (d) It has led to the failure of many small businesses.

解答と解説

問 30

正解

(b) A day of chaos and congestion for police.

解説

文章の第 2 段落に、「警察は、郊外からの買い物客や観光客が街に押し寄せたことで発生した**混乱** (chaos) を表現するために、『ブラックフライデー』という言葉を使った」と書かれています。

重要単語の解説:

- chaos (名詞) : 混乱、無秩序。
- congestion (名詞) : 混雑、渋滞。特に交通や人混みの密集を指します。

問 31

正解

(b) The police in Philadelphia.

解説

第 2 段落に、「この用語が**フィラデルフィアの警察**によって使われ始めた」と明確に述べられています。

問 32

正解

(b) By linking it to businesses making a profit.

解説

第 3 段落に、「小売業者は、その年初めて**黒字**になる (in the black) 日だと説明することで、この言葉の意味を肯定的なものに変えた」と書かれています。「黒字」とは利益が出ている状態を指します。

重要単語の解説:

- retailer (名詞) : 小売業者。
- in the red / in the black (イディオム) : 赤字 (損失) である / 黒字 (利益) である。

問 33

正解

(b) It encourages consumerism and overconsumption.

解説

最終段落に、「この世界的な拡大は、**消費主義（consumerism）**と環境への影響についての疑問も提起している」と述べられています。特に、「絶え間なく新しい商品を買うというプレッシャーが**過剰消費（overconsumption）**につながる」と指摘しています。

重要単語の解説:

- **consumerism**（名詞）：消費主義。物を大量に購入・消費することが良いことだとする考え方。
- **overconsumption**（名詞）：過剰消費。
- **consequence**（名詞）：結果、結末。特に、望ましくない結果を指すことが多いです。

0.9 宇宙開発競争

次の英文を読み、後の設問に答えなさい。

The Space Race: A Cold War Competition Beyond Earth

The Space Race was a 20th-century competition between the two Cold War rivals, the United States and the Soviet Union, for **supremacy** in spaceflight capability. Beginning in the late 1950s, this race became a highly visible and dramatic part of their broader political, military, and technological rivalry. The nations aimed to prove their technological and ideological superiority by achieving key milestones in space exploration.

The Soviet Union achieved the first major victory in 1957 by launching **Sputnik**, the world's first artificial satellite, into orbit. This event shocked the United States, which had long assumed its technological leadership. The US responded by creating the National Aeronautics and Space Administration (NASA) and accelerating its own space program. The Soviets continued their lead by sending the first human, Yuri Gagarin, into space in 1961.

However, the United States set its sights on a bigger goal: the moon. President John F. Kennedy declared in 1961 that the US would land a man on the moon before the end of the decade. This ambitious challenge, known as the Apollo program, became a national focus. Despite several setbacks, NASA successfully landed astronauts Neil Armstrong and Buzz Aldrin on the lunar surface on July 20, 1969. Armstrong's famous words, "That's one small step for a man, one giant leap for mankind," symbolized not only a scientific achievement but also a major victory for the US.

Ultimately, the Space Race was more than just a contest for rockets and satellites; it was a battle of ideologies. It drove immense technological innovation, from new materials to computing systems, many of which are now part of our daily lives. While the rivalry eventually faded, the legacy of the Space Race continues to inspire generations of scientists and engineers to look to the stars.

設問

- 問 34. What was the primary purpose of the Space Race for the United States and the Soviet Union?
- (a) To find new planets for human colonization.
 - (b) To establish military bases in space.
 - (c) To prove their technological and ideological superiority.
 - (d) To search for alien life forms.

- 問 35. What was the first major achievement of the Soviet Union in the Space Race?

- (a) Sending the first woman into space.
- (b) Landing a man on the moon.
- ☒ (c) Launching the first artificial satellite.
- (d) Building the first space station.

問 36. According to the passage, who was the first person to travel into space?

- (a) Neil Armstrong
- (b) Buzz Aldrin
- (c) John F. Kennedy
- ☒ (d) Yuri Gagarin

問 37. What was the Apollo program's main goal?

- (a) To build a satellite that could orbit Earth.
- (b) To send a spacecraft to another planet.
- ☒ (c) To land a human on the moon.
- (d) To establish a partnership with the Soviet Union.

解答と解説

問 34

正解

(c) To prove their technological and ideological superiority.

解説

文章の第1段落で、両国が「宇宙探査における主要な節目を達成することで、**技術的および思想的な優位性**を証明することを目指した」と明確に述べられています。これは単なる科学競争ではなく、政治的な競争でもあったことがわかります。

重要単語の解説:

- **supremacy** (名詞) : 至上権、優位性。
- **ideological** (形容詞) : 思想的な、イデオロギーの。

問 35

正解

(c) Launching the first artificial satellite.

解説

第2段落に、「ソビエト連邦は、世界初の人工衛星である**スプートニク**を軌道に乗せることで、最初の大きな勝利を達成した」と書かれています。

問 36

正解

(d) Yuri Gagarin

解説

第2段落の最後で、「ソ連は、初の人間である**ユーリ・ガガーリン**を1961年に宇宙に送り込むことで、そのリードを維持した」と述べられています。

問 37

正解

(c) To land a human on the moon.

解説

第3段落で、ケネディ大統領が「10年が終わるまでに**人間を月面に着陸させる**」と宣言したことが、アポロ計画という「野心的な挑戦」だったと説明されています。

重要単語の解説:

- **ambitious**（形容詞）：野心的な、意欲的な。
- **lunar**（形容詞）：月の。

0.10 小説『宇宙戦争』と社会的影響

次の英文を読み、後の設問に答えなさい。

A Broadcast of Panic: How H.G. Wells's 'The War of the Worlds' Changed Society

H.G. Wells's 1898 novel, "The War of the Worlds," is a classic of science fiction, **depicting** an invasion of Earth by Martians. While the novel itself was a popular success, its most significant social impact came much later, on October 30, 1938, when a radio broadcast of the story caused widespread panic across the United States. This event became a famous case study in media history, revealing the power of mass media and the public's **susceptibility** to misinformation.

The broadcast was an adaptation of Wells's novel, presented by Orson Welles and his Mercury Theatre on the Air. The program was formatted as a series of realistic news bulletins, with fictional reports of Martian spaceships landing, military responses, and chaos in the streets. Although the show began with a clear announcement that it was a fictional drama, many listeners tuned in late and missed the **disclaimer**. This led them to believe that a genuine alien invasion was underway.

As the broadcast continued, thousands of people called police stations and newsrooms in a state of alarm. Some even fled their homes, preparing for a real-life attack. The next day, newspapers across the country ran headlines about the panic, and the incident sparked a national conversation about the reliability of news media. It highlighted how easily a realistic presentation, combined with public anxiety about the threat of war in Europe, could lead to mass hysteria.

In the long term, the "panic broadcast" served as a powerful lesson. It demonstrated the profound influence that radio, a relatively new medium at the time, could have on public opinion and behavior. It also led to new regulations and journalistic standards aimed at preventing similar incidents. While the event is often remembered as an amusing **anecdote**, it stands as a serious reminder of the need for media literacy and critical thinking in an age of instantaneous information.

設問

問 38. What was the most significant social impact of the radio broadcast of "The War of the Worlds"?

- (a) It made Orson Welles a famous actor.
- (b) It led to a real-life alien invasion.
- (c) It caused widespread panic and highlighted the power of media.
- (d) It discouraged people from reading science fiction novels.

問 39. How was the radio program formatted to sound so realistic?

- (a) It featured real military personnel.
- (b) It was presented as a series of news bulletins.
- (c) It included live reports from the landing site.
- (d) It was broadcast from a secret government location.

問 40. What factor, besides the format of the broadcast, contributed to the panic?

- (a) The public's lack of interest in the story.
- (b) The ongoing threat of war in Europe.
- (c) The government's official confirmation of the invasion.
- (d) The widespread availability of television.

問 41. What was a long-term consequence of the panic broadcast?

- (a) The novel was banned in the United States.
- (b) People stopped listening to the radio.
- (c) It led to new regulations for media and journalism.
- (d) It proved that people were not easily fooled by fake news.

解答と解説

問 38

正解

(c) It caused widespread panic and highlighted the power of media.

解説

文章の第1段落に、「その最も重要な社会的影響は、ラジオ放送が**広範囲にわたるパニック**を引き起こし、**マスメディアの力**を明らかにしたことだ」と書かれています。

重要単語の解説:

- **widespread** (形容詞) : 広範囲に及ぶ、広く行き渡った。
- **susceptibility** (名詞) : 影響を受けやすさ、感受性。

問 39

正解

(b) It was presented as a series of news bulletins.

解説

第2段落に、「その番組は、現実的な**一連のニュース速報**として構成されていた」と述べられており、これが多くのリスナーを信じ込ませた理由です。

重要単語の解説:

- **bulletin** (名詞) : 速報、報道。
- **disclaimer** (名詞) : 免責事項、非保証声明。ここでは「フィクションであること」を告げる部分。

問 40

正解

(b) The ongoing threat of war in Europe.

解説

第3段落に、パニックの原因として「現実的な演出」に加え、「ヨーロッパでの戦争の脅威に対する**人々の不安**」が挙げられています。

重要単語の解説:

- **anxiety** (名詞) : 不安、心配。
- **mass hysteria** (名詞) : 集団ヒステリー、集団パニック。

問 41

正解

(c) It led to new regulations for media and journalism.

解説

最終段落で、この放送事件の長期的な影響として、「同様の事件を防ぐことを目的とした**新しい規制**や**ジャーナリズムの基準**につながった」と説明されています。

重要単語の解説:

- － **regulation** (名詞) : 規制、規則。
- － **journalistic** (形容詞) : ジャーナリズムの、報道の。
- － **literacy** (名詞) : 読み書き能力。ここでは、メディア情報を適切に理解・判断する能力を指す「メディアリテラシー」の意味で使われています。

0.11 文化相対主義と自民族中心主義

次の英文を読み、後の設問に答えなさい。

Understanding Others: The Conflict Between Cultural Relativism and Ethnocentrism

When we encounter cultures different from our own, our immediate reaction is often to compare their practices to ours. This tendency to judge other cultures based on the standards and values of one's own is known as ****ethnocentrism****. While it is a natural human tendency, it can lead to misunderstanding and conflict, as it assumes that one's own culture is the correct or superior way of life.

Anthropology, the study of human societies and their cultures, offers an alternative perspective: ****cultural relativism****. This principle suggests that a person's beliefs and activities should be understood by others in terms of that individual's own culture. Instead of judging, cultural relativism encourages us to appreciate that different cultures have their own unique logic and context. For example, a marriage ritual that seems strange to an outsider may hold deep historical and spiritual meaning for those within that culture.

However, cultural relativism is not without its **critics**. Opponents argue that if taken to an extreme, it can be used to justify practices that violate universal human rights, such as slavery or gender inequality. They question whether it is possible, or even right, to view all cultural practices as equally valid, especially when they cause harm to individuals. This debate highlights the central challenge of cultural relativism: how to respect cultural differences without abandoning our ethical principles.

In conclusion, both ethnocentrism and cultural relativism present complex challenges. While ethnocentrism can lead to prejudice and misunderstandings, cultural relativism requires careful consideration to avoid moral dilemmas. The key is to find a balance—to maintain an open mind and seek to understand different cultures from their own perspective, while still upholding fundamental human values.

設問

問 42. What is the main difference between ethnocentrism and cultural relativism?

- (a) Ethnocentrism is a principle used by anthropologists, while cultural relativism is a common human tendency.
- (b) Ethnocentrism leads to understanding other cultures, while cultural relativism leads to conflict.
- (c) Ethnocentrism judges other cultures by one's own standards, while cultural relativism seeks

to understand them within their own context.

- (d) Ethnocentrism promotes global peace, while cultural relativism promotes cultural separation.

問 43. According to the passage, what is a potential negative outcome of ethnocentrism?

- (a) It leads to a deeper appreciation of one's own culture.
(b) It can lead to misunderstandings and conflict.
(c) It helps people learn new languages.
(d) It encourages the study of anthropology.

問 44. What is a major criticism of cultural relativism mentioned in the passage?

- (a) It is difficult to apply to daily life.
(b) It can be used to justify harmful practices.
(c) It is not a widely accepted idea.
(d) It prevents people from expressing their own values.

問 45. What does the author suggest as a solution to the conflict between the two concepts?

- (a) To completely abandon cultural relativism.
(b) To always prioritize one's own cultural values.
(c) To find a balance between understanding and upholding ethical principles.
(d) To ignore all cultural differences and focus on similarities.

解答と解説

問 42

正解

(c) Ethnocentrism judges other cultures by one's own standards, while cultural relativism seeks to understand them within their own context.

解説

文章の第1段落と第2段落で、両者の定義が対比されています。**ethnocentrism**は「自文化の基準に基づいて他文化を判断する傾向」であり、一方の**cultural relativism**は「個人の文化の観点から信念や行動を理解する」という原則だと説明されています。

重要単語の解説:

- **ethnocentrism** (名詞): 自民族中心主義。自分の文化が最も優れていると見なす考え方。
- **cultural relativism** (名詞): 文化相対主義。文化には優劣がなく、それぞれを固有の文脈で理解すべきだという考え方。

問 43

正解

(b) It can lead to misunderstandings and conflict.

解説

第1段落で、「**ethnocentrism**は、自分の文化が正しいという前提を持つため、**誤解や対立**につながる可能性がある」と明確に述べられています。

問 44

正解

(b) It can be used to justify harmful practices.

解説

第3段落に、「反対派は、極端な**cultural relativism**は、奴隷制度やジェンダー不平等といった、普遍的な人権を侵害する行為を**正当化**するために使われかねないと主張する」と書かれています。

重要単語の解説:

- **justify** (動詞): 正当化する。
- **violate** (動詞): 違反する、侵害する。
- **universal human rights** (名詞): 普遍的人権。

問 45

正解

(c) To find a balance between understanding and upholding ethical principles.

解説

文章の最後の文で、筆者は「鍵となるのは**バランスを見つけること**だ」と結論づけています。具体的には、「異なる文化を彼ら自身の視点から理解しようと努めつつ、**基本的な人間の価値観を堅持すること**」だと述べられています。

重要単語の解説:

- **uphold** (動詞) : 堅持する、守る。
- **fundamental** (形容詞) : 根本的な、基本的な。

✧ 0.12 ナスカの地上絵

次の英文を読み、後の設問に答えなさい。

The Mysterious Nazca Lines: Art on a Grand Scale

In the **arid plains** of Peru lie the Nazca Lines, a series of ancient geoglyphs etched into the desert floor. These monumental drawings, some stretching over 200 meters long, **depict** a wide variety of figures, including animals like hummingbirds, monkeys, and spiders, as well as plants and geometric shapes. Created by the Nazca culture between 500 B.C. and A.D. 500, they are so vast that they can only be fully appreciated from high in the air.

The creation of the lines was a relatively simple process. The Nazca people removed the **reddish-brown** iron oxide **pebbles** that cover the desert surface, revealing the lighter, contrasting sand underneath. However, the purpose of these geoglyphs has remained a profound mystery for archaeologists and historians. Theories range from astronomical calendars used to track the seasons to ritual pathways for religious ceremonies. Others suggest they were messages to the gods, visible from the heavens.

Despite their age, the Nazca Lines have survived for centuries due to the region's unique climate. The Nazca Desert is one of the **driest** places on Earth, with virtually no rain or wind to erode the lines. However, modern threats, such as illegal mining, new highways, and climate change, pose serious risks to their preservation. The delicate nature of the lines means they are easily damaged and difficult to restore.

Today, efforts are being made to protect this UNESCO World Heritage site. Researchers use drones and satellite imagery to monitor the area for damage, and the Peruvian government has implemented strict regulations to control access to the site. While the original meaning of the lines may never be known for certain, preserving them ensures that this remarkable piece of human history can continue to inspire wonder and study for generations to come.

設問

問 46. What is the main idea of the first paragraph?

- (a) The theories about why the Nazca Lines were created.
- (b) The process of how the Nazca Lines were made.
- (c) A physical description of the Nazca Lines and how they are viewed.) The history of the Nazca civilization and its daily life.

問 47. According to the passage, what has protected the Nazca Lines from natural erosion for centuries?

- (a) The Nazca people's advanced construction techniques.
- ☒ (b) The arid and stable climate of the Nazca Desert.
- ☒ (c) The government's protection efforts.
- (d) The lines being hidden beneath the sand.

問 48. Based on the passage, which of the following is NOT a modern threat to the Nazca Lines?

- (a) Illegal mining.
- (b) New highways.
- (c) Climate change.
- ☒ (d) Strong winds and heavy rain.

問 49. What is the primary focus of modern preservation efforts for the Nazca Lines?

- (a) Building a museum to display the lines.
- (b) Restoring the damaged parts of the lines.
- ☒ (c) Using advanced technology to monitor and control access to the site.
- (d) Solving the mystery of their original purpose.

解答と解説

問 46

正解

(c) A physical description of the Nazca Lines and how they are viewed.

解説

第1段落では、ナスカの地上絵が「ペルーの乾燥した平原にある**古代の地上絵**」であること、その**大きさ**や**描かれているもの**（ハチドリ、サルなど）、そして「**上空からでなければ全体像が見えない**」という見方について説明しています。

重要単語の解説:

- **arid** (形容詞) : 乾燥した、不毛な。
- **geoglyph** (名詞) : 地上絵。地面に描かれた巨大な絵や模様。
- **etch** (動詞) : 刻み込む、エッチングする。

問 47

正解

(b) The arid and stable climate of the Nazca Desert.

解説

第3段落に、「ナスカの地上絵が何世紀にもわたって生き残ってきたのは、この地域のユニークな気候のおかげだ」と書かれています。続けて、「ナスカ砂漠は地球上で最も乾燥した場所の一つであり、地上絵を浸食する**雨や風がほとんどない**」と説明されています。

重要単語の解説:

- **erosion** (名詞) : 浸食、腐食。
- **delicate** (形容詞) : 繊細な、もろい。

問 48

正解

(d) Strong winds and heavy rain.

解説

第3段落に、「違法な採掘、新しい高速道路、気候変動」が現代の脅威として挙げられています。一方、(d)「強風と豪雨」は、この地域の**気候特性として地上絵の保存を助けている要因**であり、脅威ではありません。

問 49

正解

(c) Using advanced technology to monitor and control access to the site.

解説

最終段落に、「研究者は**ドローンや衛星画像**を使って被害を監視し、ペルー政府は立ち入りを規制するための**厳しい規則**を施行している」と書かれており、これらが主な保存の取り組みであることがわかります。

重要単語の解説:

- － **preservation** (名詞) : 保存、保全。
- － **monitor** (動詞) : 監視する。
- － **regulate** (動詞) : 規制する。

0.13 インカ帝国

次の英文を読み、後の設問に答えなさい。

The Empire of the Sun: Rise and Fall of the Inca Civilization

The Inca Empire, or **Tawantinsuyu**, was a remarkable civilization that flourished in the Andean region of South America from the early 13th century until the 16th century. Stretching over 4,000 kilometers, it was the largest empire in pre-Columbian America. Despite having no written language or a formal system of currency, the Inca built a vast and highly organized society. They achieved incredible **feats** of engineering, creating an extensive network of roads and **majestic** stone structures like the fortress of Machu Picchu, which remains a marvel of construction.

The success of the Inca Empire was based on a system of collective labor called the *mit'a*. Rather than paying taxes in money, citizens contributed their labor to state projects, such as building roads, bridges, and agricultural terraces. The government, in turn, provided food and supplies for its people. This system ensured the efficient distribution of resources throughout the empire and allowed for large-scale public works. The Inca also used a complex system of **knotted** strings, known as *quipu*, for record-keeping, which helped them manage their vast population and resources.

However, the empire's impressive reign came to a sudden and tragic end with the arrival of the Spanish conquistadors, led by Francisco Pizarro, in the 1530s. The Spanish, though vastly outnumbered, possessed superior military technology, including steel weapons and firearms. The Inca Empire was also weakened by internal conflict and the spread of **smallpox**, a disease brought by the Europeans to which the Inca had no *immunity*. This combination of factors led to the swift collapse of a powerful civilization and the end of a unique way of life.

Today, the legacy of the Inca lives on in the indigenous communities of the Andes and in the awe-inspiring ruins they left behind. Studying their civilization offers us profound lessons about human ingenuity, social organization, and the devastating impact of external forces on a culture.

設問

問 50. What is a key characteristic of the Inca Empire mentioned in the first paragraph?

- (a) They had a sophisticated written language.
- (b) They used a formal system of currency for trade.
- (c) They built an extensive network of roads.
- (d) They were the smallest empire in pre-Columbian America.

問 51. How did the *mit'a* system work in the Inca Empire?

- (a) Citizens paid taxes in money to the government.
- (b) Citizens contributed their labor to public projects instead of paying taxes.
- (c) The government used a system of currency to distribute resources.
- (d) Foreigners were required to work on construction projects.

問 52. According to the passage, which of the following was a factor that contributed to the fall of the Inca Empire?

- (a) The Inca military's superior technology.
- (b) The empire's strong central government.
- (c) The lack of internal conflict within the empire.
- (d) The spread of a European disease.

問 53. What is the main purpose of the passage's final paragraph?

- (a) To describe the beauty of the Inca ruins.
- (b) To explain how the Inca Empire was rediscovered.
- (c) To summarize the Inca's influence today and the lessons we can learn from them.
- (d) To predict the future of indigenous communities in the Andes.

解答と解説

問 50

正解

(c) They built an extensive network of roads.

解説

第1段落に「彼らは**広大な道路網**やマチュピチュのような壮大な石造建築物という、信じられないほどの工学的な偉業を成し遂げた」と書かれています。

重要単語の解説:

flourish (繁栄する、栄える)

pre-Columbian (コロンブス以前の)

feat (偉業、功績)

majestic (雄大な、荘厳な)

問 51

正解

(b) Citizens contributed their labor to public projects instead of paying taxes.

解説

第2段落に「市民は金銭で税金を払う代わりに、道路や橋、農地の段々畑を造るような国家事業に**労働を拠出**した」と説明されています。これが**mit'a** (ミータ) という制度です。

重要単語の解説:

collective (集団的な、共同の)

contribute (貢献する、拠出する)

territory (領土、領域)

問 52

正解

(d) The spread of a European disease.

解説

第3段落に、「インカ帝国は**内部紛争**と、ヨーロッパ人が持ち込んだ**天然痘**という病気の蔓延によっても弱体化していた」と述べられています。

重要単語の解説:

conquistador (コンキスタドール (スペインの征服者))

outnumbered (数で勝る、数が上回る)

firearm (火器、銃)

smallpox (天然痘)

immunity (免疫)

問 53

正解

(c) To summarize the Inca's influence today and the lessons we can learn from them.

解説

最終段落では、インカの「遺産は今も生きている」と述べ、彼らの文明を学ぶことは「**人類の創意工夫**、**社会組織**、そして**外部からの力による壊滅的な影響**について、私たちに**深い教訓**を与えてくれる」と結論づけています。

重要単語の解説:

legacy (遺産、遺物)

indigenous (土着の、先住の)

ingenuity (創意工夫、巧妙さ)

devastating (壊滅的な、甚大な)

0.14 大航海時代と対数の歴史

次の英文を読み、後の設問に答えなさい。

Navigating the Seas: How the Age of Exploration Drove Mathematical Innovation

At first glance, the Age of Exploration and the history of mathematics seem to have little in common. The former was a period of daring voyages and new discoveries, while the **latter** is an abstract science of numbers and equations. However, one of the most significant mathematical innovations, the logarithm, was directly born from the practical needs of navigators and astronomers during this era.

Before the invention of logarithms, sailors and astronomers had to perform incredibly complex and time-consuming calculations to determine their position at sea. These calculations involved multiplying and dividing very large numbers, which was prone to human error. A small mistake could have devastating consequences, leading a ship far off course or to disaster. The need for a faster and more accurate method was a major ****incentive**** for mathematicians to find a solution.

In the late 16th century, Scottish mathematician John Napier invented logarithms. A logarithm is essentially an exponent: it's the power to which a number must be raised to produce another number. For example, the logarithm of 100 to the base 10 is 2, because 10 raised to the power of 2 is 100 ($10^2 = 100$). The genius of logarithms is that they transform multiplication and division problems into much simpler addition and subtraction problems. For navigators, this meant that multiplying two large numbers, a task that might take hours, could be reduced to looking up their logarithms in a table and simply adding them together.

The impact of logarithms was immediate and profound. They dramatically simplified the calculations required for navigation, astronomy, and other fields of science, saving countless hours and lives. The relationship between the Age of Exploration and the invention of logarithms serves as a powerful example of how real-world problems can drive scientific and mathematical progress.

設問

問 54. What problem did navigators and astronomers face before the invention of logarithms?

- (a) They could not measure distance at all.
- (b) Their calculations were too fast and led to errors.
- (c) They had to perform complex and time-consuming calculations.
- (d) They lacked the proper tools for measuring the stars.

$\times, \div$
 $\downarrow$ transform
 $+, -$

- 問 55. According to the second paragraph, what does the author suggest was a major ****incentive**** for mathematicians to invent logarithms?
- (a) The desire to make mathematics more abstract.
 - (b) The need to find a faster and more accurate way to navigate.
 - (c) The pleasure of solving a difficult mathematical puzzle.
 - (d) The curiosity about how ships travel on the sea.
- 問 56. What is the main benefit of using logarithms, as described in the passage?
- (a) They allow for the quick and easy conversion of addition into multiplication.
 - (b) They are used to simplify addition problems into subtraction. ✕
 - (c) They are a more efficient way to perform division and multiplication.
 - (d) They completely eliminated the need for any calculations at sea.
- 問 57. What does the passage say about the relationship between the Age of Exploration and logarithms?
- (a) They were two unrelated historical developments.
 - (b) The Age of Exploration was made possible by the invention of logarithms.
 - (c) Logarithms were invented as a direct result of the needs of the Age of Exploration.
 - (d) The invention of logarithms led to a decline in sea travel.

解答と解説

問 54

正解

(c) They had to perform complex and time-consuming calculations.

解説

第2段落冒頭に「対数の発明前、航海士や天文学者は、海上で自分の位置を特定するために**信じられないほど複雑で時間のかかる計算**をしなければならなかった」と書かれています。

重要単語の解説:

navigators (航海士、ナビゲーター)

astronomers (天文学者)

time-consuming (時間のかかる、手間のかかる)

問 55

正解

(b) The need to find a faster and more accurate way to navigate.

解説

第2段落の最後で、「より速く、より正確な方法の必要性が、数学者たちが解決策を見つけるための**大きな動機**となった」と述べられています。

重要単語の解説:

incentive (動機、誘因)

prone to (～しがちである、～の傾向がある)

devastating (破壊的な、壊滅的な)

問 56

正解

(c) They are a more efficient way to perform division and multiplication.

解説

第3段落に、「対数の天才的な点は、掛け算や割り算の問題を、**はるかに単純な足し算や引き算の問題に変える**ことだ」と説明されています。

重要単語の解説:

transform (変形させる、変換する)

genius (天才性、非凡な才能)

look up ((辞書などで) 調べる)

問 57

正解

(c) Logarithms were invented as a direct result of the needs of the Age of Exploration.

解説

文章全体を通して、対数の発明は、大航海時代における航海や天文学の**実用的な必要性から直接生まれた**ことが強調されています。第1段落では「対数という最も重要な数学的革新の一つは、この時代の航海士や天文学者の**実用的なニーズから直接生まれた**」と述べられています。

重要単語の解説:

innovation (革新、イノベーション)

profound (深遠な、重大な)

progress (進歩、発展)

0.15 超伝導とリニアモーターカー

次の英文を読み、後の設問に答えなさい。

Defying Gravity: The Science of Superconductivity and Maglev Trains

Superconductivity is a fascinating phenomenon in physics where a material, when cooled below a certain temperature, loses all electrical resistance. This means that an electric current can flow through it forever without losing any energy. Another remarkable property of superconductors is that they ****expel**** magnetic fields. This effect, known as the Meissner effect, causes a superconductor to **levitate** when placed near a powerful magnet. It is this unique combination of zero resistance and magnetic **levitation** that makes superconductivity a key technology for the future.

The most well-known application of superconductivity is in Maglev (Magnetic Levitation) trains. Unlike conventional trains that run on wheels, Maglev trains are suspended above the track by powerful magnetic forces. The train cars are equipped with superconducting magnets, and the tracks contain a series of coils. When the superconducting magnets are cooled to their operating temperature, they create a strong magnetic field. As this field interacts with the coils on the track, the train is lifted and **propelled** forward with incredible speed.

One of the biggest advantages of Maglev trains is their speed. Because there is no friction from wheels on a track, they can travel at much higher speeds than traditional trains, often exceeding 600 kilometers per hour. This also means the ride is extremely smooth and quiet. While the technology holds great promise, its widespread use is currently limited by the high cost of the specialized materials and the complex infrastructure required for the tracks. Researchers are working to develop new superconducting materials that can operate at higher temperatures, which would significantly reduce the cooling costs and make the technology more accessible.

In conclusion, superconductivity is not just a scientific curiosity; it is a powerful principle that has enabled **groundbreaking** innovations like the Maglev train. As we continue to advance our understanding of this phenomenon, we can expect to see it applied to new technologies that will transform the way we live and travel.

設問

問 58. What are the two main properties of a superconductor mentioned in the first paragraph?

- (a) They produce heat and attract magnetic fields.
- (b) They have zero electrical resistance and expel magnetic fields.

- (c) They are very heavy and absorb electricity.
- (d) They are strong and resistant to high temperatures.

問 59. According to the passage, how is the Maglev train lifted above the track?

- (a) By strong air pressure from below the train.
- (b) By a special type of wheel and rail system.
- (c) By the interaction between superconducting magnets and track coils.
- (d) By the Meissner effect alone.

問 60. What is the closest meaning of the underlined word expel as it is used in the first paragraph?

- (a) to absorb or take in
- (b) to push or force out
- (c) to create or generate
- (d) to hold or contain

問 61. What is a major challenge that prevents the widespread use of Maglev trains?

- (a) Their very slow speed compared to regular trains.
- (b) The high cost of materials and infrastructure.
- (c) The loud noise they produce while traveling.
- (d) The inability to travel in different weather conditions.

解答と解説

問 58

正解

(b) They have zero electrical resistance and expel magnetic fields.

解説

第1段落に、「超伝導とは、ある物質が特定の温度以下に冷やされると、**すべての電気抵抗を失う**という物理現象である」と述べられています。また、「超伝導体のもう一つの特筆すべき特性は、**磁場を排斥する**ことである」と説明されています。

重要単語の解説:

phenomenon (現象)

resistance (抵抗)

expel (追い出す、排斥する)

levitate (浮揚する、宙に浮く)

問 59

正解

(c) By the interaction between superconducting magnets and track coils.

解説

第2段落に、「超伝導磁石が線路上のコイルと**相互作用する**ことで、列車は持ち上げられ、驚異的な速度で前方に推進される」と述べられています。

重要単語の解説:

conventional (従来の、一般的な)

suspend (吊り下げる、宙に浮かせる)

coil (コイル)

propel (推進する、前進させる)

問 60

正解

(b) to push or force out

解説

expel (動詞) は、「追い出す、排斥する」という意味です。この文脈では、超伝導体が「磁場を外に押し出す」という性質を表すために使われています。

重要単語の解説:

expel (追い出す、排斥する)

Meissner effect (マイスナー効果)：超伝導体が磁場を外部に押し出す現象。

問 61

正解

(b) The high cost of materials and infrastructure.

解説

第3段落に、「その普及は、**特殊な材料の高コスト**と、線路に必要な**複雑なインフラ**によって現在制限されている」と明確に書かれています。

重要単語の解説:

widespread (広範囲にわたる、普及した)

infrastructure (インフラ、社会基盤)

accessible (利用しやすい、入手しやすい)

0.16 日米の高校生活の対比

次の英文を読み、後の設問に答えなさい。

High School Life: A Tale of Two Systems

High school experiences can vary greatly depending on where you live. While both Japanese and American high schools aim to educate young people, the day-to-day life and underlying philosophies of these two systems are quite different. Understanding these differences can offer a fascinating glimpse into the cultural values of each society.

One major distinction lies in the academic structure. In many Japanese high schools, students typically stay in the same classroom for most of their classes, with teachers rotating in and out. The curriculum is often standardized across the country, with a strong focus on preparing students for **rigorous** entrance exams. In contrast, American high school students move from classroom to classroom for each subject. The curriculum offers a wider range of elective courses, allowing students to ****specialize**** in areas of interest and explore subjects like journalism, drama, or computer science.

Another key difference is the role of extracurricular activities. In Japan, club activities (*bukatsu*) are a highly integrated part of school life, often requiring significant time and dedication from students after school. These clubs, whether for sports or culture, are seen as a way to build character and teamwork. In the U.S., extracurriculars are equally important, but they often have a strong connection to college admissions. Students are encouraged to participate in a variety of activities to demonstrate their talents and leadership skills, which can make their college applications more ****competitive****.

Finally, the social environment differs. While Japanese high schools often emphasize group harmony and conformity, American high schools tend to encourage individualism and self-expression. This is reflected in everything from school uniforms (common in Japan but rare in the U.S.) to classroom discussions. Neither system is inherently superior; each reflects the unique priorities and values of its culture.

設問

問 62. What is the main idea of the passage?

- (a) American high schools are better than Japanese high schools.
- (b) The differences in the academic and social systems of high schools in Japan and the U.S.
- (c) The history of high school education in both Japan and the U.S.
- (d) The importance of taking elective courses in high school.

問 63. According to the passage, how do Japanese and American high school curricula differ?

- (a) Japanese schools offer more elective courses.
- (b) American schools focus more on preparing for entrance exams.
- ☒ (c) Japanese schools have a more standardized curriculum.
- (d) American schools emphasize group harmony and conformity.

問 64. In the second paragraph, what is the closest meaning of the underlined word specialize?

- (a) to gain a comprehensive understanding of many subjects
- (b) to become a professional in a specific field
- ☒ (c) to focus on a particular area of study
- (d) to take only the most difficult classes

問 65. What is a key difference in the social environment between Japanese and American high schools?

- (a) Japanese schools prioritize individual expression.
- (b) American schools emphasize group harmony.
- ☒ (c) Japanese schools often use school uniforms.
- (d) American schools do not have extracurricular activities.

解答と解説

問 62

正解

(b) The differences in the academic and social systems of high schools in Japan and the U.S.

解説

文章全体を通して、日本の高校とアメリカの高校の学業、課外活動、社会的環境といった様々な側面における**違い**に焦点を当てて説明しています。

重要単語の解説:

underlying (根底にある、根本的な)

philosophy (哲学、考え方)

glimpse (ちらりと見ること)

問 63

正解

(c) Japanese schools have a more standardized curriculum.

解説

第2段落に、「日本の高校では**カリキュラムが全国的に標準化されている**ことが多い」と書かれています。一方、アメリカの学校では「より幅広い**選択科目を**提供している」と述べられています。

重要単語の解説:

academic (学問的な、学業の)

curriculum (カリキュラム、教育課程)

standardized (標準化された)

rigorous (厳格な、厳しい)

問 64

正解

(c) to focus on a particular area of study

解説

specialize (動詞) は、「ある分野に特化する、専門とする」という意味です。この文脈では、生徒が「興味のある分野に**集中する**ことを可能にする」という意味で使われています。

重要単語の解説:

specialize (専門とする、特化する)

elective (選択科目)

問 65

正解

(c) Japanese schools often use school uniforms.

解説

最終段落に、「この違いは、**制服**（日本では一般的だが、米国では珍しい）から教室での議論に至るまで、あらゆるものに反映されている」と書かれており、(c) は事実です。また、日本の高校では**集団の調和**と**協調性**が重視されるのに対し、アメリカでは**個人主義**と**自己表現**が奨励されるという違いも述べられています。

重要単語の解説:

harmony (調和)

conformity (協調性、順応性)

individualism (個人主義)

self-expression (自己表現)

0.17 大学の教育システム

次の英文を読み、後の設問に答えなさい。

Beyond the Major: The Value of Liberal Arts Education

When choosing a university, a student often has to decide between two main educational systems: a liberal arts approach or a more specialized, professional one. While both systems prepare students for a career, they do so with different philosophies about what a well-rounded education entails. Understanding these philosophies can help students make a more informed decision about their future.

A liberal arts education, common in many universities, is designed to provide a broad knowledge base across various disciplines. Students are encouraged to take classes in the humanities, social sciences, and natural sciences, in addition to their chosen major. This **holistic** approach aims to develop critical thinking, communication, and problem-solving skills, preparing students not just for a specific job but for a dynamic and changing world. The idea is that a wide range of knowledge makes a person more adaptable and capable of tackling complex, interdisciplinary challenges.

In contrast, a specialized, professional education focuses on providing deep, focused knowledge and practical skills for a particular field, such as engineering, medicine, or business. The curriculum is often highly structured, with little room for elective courses outside of the core requirements. This system is efficient for students who are certain about their career path and want to acquire the specific expertise needed to enter their chosen profession directly.

While critics of a liberal arts education sometimes argue that it is not practical enough for the modern job market, its supporters emphasize its long-term value. They claim that employers are increasingly looking for employees who can think creatively, communicate effectively, and collaborate with others—skills that are central to the liberal arts philosophy. As technology automates more routine tasks, these uniquely human skills will become even more valuable. The debate over which system is "better" will likely continue, but both offer distinct paths to a successful and fulfilling life.

設問

問 66. What is the main purpose of this passage?

- (a) To argue that a liberal arts education is superior to a professional one.
- (b) To explain the differences between liberal arts and specialized education.
- (c) To describe the specific curriculum of a single university.
- (d) To convince students to choose a professional education.

問 67. According to the second paragraph, what is a key goal of a liberal arts education?

- (a) To train students for a single, specific career.
- ☒ (b) To provide a broad knowledge base and develop transferable skills.
- (c) To allow students to take only classes in their major.
- (d) To focus exclusively on scientific disciplines.

問 68. In the second paragraph, what is the closest meaning of the underlined word holistic?

- (a) focused on a single, specific area
- (b) practical and geared towards a job
- ☒ (c) concerned with the entire person or system
- (d) narrow and detailed

問 69. What is an argument for a liberal arts education's long-term value, as mentioned in the final paragraph?

- (a) It teaches practical, technical skills that are always in demand.
- (b) It helps students enter a specific profession directly after graduation.
- ☒ (c) It develops uniquely human skills like creative thinking and collaboration.
- (d) It is more affordable than a professional education.

解答と解説

問 66

正解

(b) To explain the differences between liberal arts and specialized education.

解説

この文章は、リベラルアーツ教育と専門教育という二つの教育システムを比較し、それぞれの特徴と哲学を説明しています。

重要単語の解説:

entail (～を伴う、～を必要とする)

informed (情報に基づいた、十分な知識を持った)

問 67

正解

(b) To provide a broad knowledge base and develop transferable skills.

解説

第2段落に、「リベラルアーツ教育は、様々な分野にわたる**幅広い知識基盤**を提供することを目的としている」とあります。さらに、「特定の仕事のためだけでなく、**批判的思考やコミュニケーションスキル**などを発展させる」と述べられています。これらのスキルは、どのような分野にも応用できるため、「transferable skills (応用可能なスキル)」と言い換えられます。

重要単語の解説:

holistic (全体的な、包括的な)

discipline (学問分野)

transferable skills (応用可能なスキル)

問 68

正解

(c) concerned with the entire person or system

解説

holistic (形容詞) は、「全体論的な、包括的な」という意味です。教育の文脈では、特定の分野だけでなく、学生の「**全体**」的な成長を目的とするアプローチを指します。

重要単語の解説:

holistic (全体的な、包括的な)

問 69

正解

(c) It develops uniquely human skills like creative thinking and collaboration.

解説

最終段落に、「リベラルアーツの支持者は、雇用主が**創造的な思考、効果的なコミュニケーション、他者との協働**といった、人間に特有のスキルを持つ人材をますます求めている」と主張しています。

重要単語の解説:

critics (批判者)

geared towards (～向けに作られた)

collaborate (協力する、協働する)

0.18 ローマ帝国の衰退

次の英文を読み、後の設問に答えなさい。

The Fall of an Empire: **Unraveling** the Decline of Rome

解明する

For centuries, the Roman Empire was the most dominant force in the Western world, a symbol of power and advanced civilization. However, by the 5th century CE, this once mighty empire had collapsed. Historians have debated the causes of its decline for centuries, and today, they agree that there was no single reason, but rather a complex combination of internal and external factors that led to its ****demise****.

One key internal factor was political instability. The empire suffered from a series of weak and short-reigned emperors, which led to civil wars and a breakdown of central authority. This instability was worsened by a severe economic crisis. High taxes were needed to support the army and the massive **bureaucracy**, but this placed a heavy burden on citizens. Trade routes became less secure, and inflation soared, further weakening the empire's financial foundations.

官僚制度

External pressures also played a crucial role. Rome's vast borders were constantly under threat from barbarian tribes such as the Goths and Vandals. The empire spent enormous resources on its military to defend these borders, but the sheer size of the territory made it impossible to protect every frontier. The increasing number of invasions from these groups put immense strain on the military and ultimately led to the loss of key territories.

In addition to these military and economic issues, social and cultural changes were occurring. As the empire grew, it became increasingly diverse, but this also led to internal divisions and a loss of civic **virtue**. Many historians also point to the rise of Christianity as a contributing factor, arguing that it shifted the focus from loyalty to the state to loyalty to the church. In conclusion, the fall of Rome was not a sudden event but a long process of decline caused by a combination of political, economic, military, and social forces.

美德

設問

問 70. What is the main argument of the passage regarding the fall of the Roman Empire?

- (a) It was caused by a single, major event.
- (b) It was primarily due to military defeats.
- (c) It was the result of a complex combination of factors.
- (d) It was an inevitable outcome of its large size.

問 71. According to the passage, which of the following was an internal political factor in Rome's decline?

- (a) Barbarian invasions.
- (b) The rise of Christianity.
- (c) A series of weak emperors.
- (d) A severe economic crisis.

問 72. In the first paragraph, what is the closest meaning of the underlined word demise?

- (a) rise
- (b) downfall
- (c) prosperity
- (d) expansion

問 73. Which of the following is mentioned as an external pressure on the Roman Empire?

- (a) High taxes.
- (b) The loss of civic virtue.
- (c) Civil wars.
- (d) Invasions from barbarian tribes.

解答と解説

問 70

正解

(c) It was the result of a complex combination of factors.

解説

第1段落で、「歴史家たちは... その衰退の単一の理由はない... むしろ**複雑な要因の組み合わせ**がその終焉につながったことに同意している」と述べられています。

重要単語の解説:

dominant (支配的な)

decline (衰退、衰え)

complex (複雑な)

combination (組み合わせ)

internal (内部の)

external (外部の)

問 71

正解

(c) A series of weak emperors.

解説

第2段落に、「一つの主要な**内部的要因**は**政治的不安定さ**であった」とあり、その具体例として「**弱く、短命の皇帝が続いた**」ことが挙げられています。

重要単語の解説:

instability (不安定さ)

central authority (中央権力)

bureaucracy (官僚制度)

問 72

正解

(b) downfall

解説

demise (名詞) は「死、終焉、崩壊」を意味します。文脈では、強大な帝国が最終的に迎えた「**終焉**」や「**崩壊**」を指しています。

重要単語の解説:

demise (終焉、崩壊)

問 73

正解

(d) Invasions from barbarian tribes.

解説

第 3 段落の冒頭に、「**外部からの圧力**もまた重要な役割を果たした」とあり、その具体的な例として「ゴート族やヴァンダル族のような**異民族の侵略**」が挙げられています。

重要単語の解説:

barbarian (異民族の、未開の)

frontier (国境、辺境)

0.19 日本の高度経済成長

次の英文を読み、後の設問に答えなさい。

Japan's Economic Miracle: The Era of Rapid Growth

Following the devastation of World War II, Japan experienced a period of unprecedented economic growth from the mid-1950s to the early 1970s. This era, often called the “Japanese Economic Miracle,” transformed the nation from a war-torn country into a major global economic power. The factors behind this rapid ****recovery**** and expansion were numerous, but so were the challenges that **arose** as a result.

起る

One of the primary drivers of this growth was a strong focus on industrial production and exports. The government and private sector worked in close collaboration, investing heavily in key industries such as steel, shipbuilding, and automobiles. Japanese companies quickly adopted and improved upon Western technologies, producing high-quality goods at competitive prices. This period saw the rise of major corporations like Toyota and Sony, which became household names around the world.

However, this rapid industrialization came at a significant cost. The intense focus on economic expansion often led to severe environmental pollution. Factories released toxic waste into rivers and the air, causing serious health problems for local residents. Furthermore, the rapid migration of people from rural areas to cities created a host of social issues, including overcrowding, inadequate housing, and the breakdown of traditional community ties. This economic boom was also criticized for leading to a sense of national materialism, as people became more focused on acquiring consumer goods than on spiritual or cultural values.

In conclusion, Japan's economic miracle was a truly remarkable period in its history, demonstrating the nation's resilience and capacity for innovation. While it successfully elevated the country's economic status and improved the living standards of many, it also served as a valuable lesson in the importance of balancing economic progress with environmental protection and social well-being.

設問

問 74. What is the main idea of the passage?

- (a) The history of Toyota and Sony's rise to global fame.
- (b) The positive and negative consequences of Japan's post-war economic growth.
- (c) The reasons why Japan became a major global exporter.
- (d) The environmental problems caused by industrialization.

問 75. According to the passage, what was a key factor in Japan's economic growth?

- (a) A focus on importing goods from Western countries.
- ☒ (b) The close collaboration between the government and private sector.
- (c) A lack of interest in industrial production.
- (d) The migration of people from cities to rural areas.

問 76. In the first paragraph, what is the closest meaning of the underlined word ***recovery***?

- (a) the process of getting worse
- (b) the act of finding something that was lost
- ☒ (c) the return to a normal or better state
- (d) the act of expanding to a larger size

問 77. Which of the following was a negative consequence of the economic boom, as mentioned in the passage?

- (a) A decrease in consumer materialism.
- (b) Improved living standards for local residents.
- (c) A decline in urban populations.
- ☒ (d) Severe environmental pollution.

解答と解説

問 74

正解

(b) The positive and negative consequences of Japan's post-war economic growth.

解説

この文章は、日本の高度経済成長がもたらした「光」の部分（経済の回復、産業の発展）と「影」の部分（環境汚染、社会問題）の両方をバランス良く論じています。

重要単語の解説:

unprecedented (前例のない)

transform (～を変形させる、～を一変させる)

numerous (多数の)

問 75

正解

(b) The close collaboration between the government and private sector.

解説

第2段落に、「政府と民間部門は**緊密に連携**して、鉄鋼、造船、自動車といった主要産業に多額の投資を行った」と書かれています。

重要単語の解説:

driver (推進力、原動力)

collaboration (協力、協業)

competitive (競争力のある)

問 76

正解

(c) the return to a normal or better state

解説

****recovery**** (名詞) は「回復、立ち直り」を意味します。文脈では、戦後の荒廃した状態から「**正常な、あるいはより良い状態に戻る過程**」を指しています。

重要単語の解説:

recovery (回復)

devastation (荒廃、破壊)

問 77

正解

(d) Severe environmental pollution.

解説

第3段落に、「この急速な工業化は重大な代償を伴った。(中略) 深刻な**環境汚染**を引き起こした」と明確に述べられています。vspace1em

重要単語の解説:**industrialization** (工業化)**significant** (重大な、重要な)**urban** (都市の) **inadequate** (不十分な、貧弱な)

0.20 日本の四大公害病

次の英文を読み、後の設問に答えなさい。

A Hidden Cost of Progress: Japan's Four Major Pollution Diseases

While Japan's period of high-speed economic growth brought about remarkable prosperity, it also left a dark legacy. The rapid and unregulated industrialization led to a series of devastating environmental disasters, known as the "Four Major Pollution Diseases." These diseases, caused by toxic industrial waste, highlighted the urgent need to balance economic goals with public health and environmental protection.

One of the most tragic cases was Minamata disease, which was caused by mercury poisoning. A chemical factory's wastewater, contaminated with a large amount of mercury, was discharged into Minamata Bay. This mercury accumulated in fish and shellfish, which were then consumed by local residents. The poison led to severe **neurological** disorders, affecting thousands of people and leaving many with **permanent** disabilities. The company's delayed response and the government's lack of action initially made the ****plight**** of the victims even worse.

Another example is Itai-itai disease, caused by cadmium poisoning. A mining company's activities led to cadmium pollution in a river in Toyama Prefecture. The contaminated water was used for irrigating rice paddies, and the cadmium was absorbed by the rice, a **staple food** for the community. This poisoning resulted in severe bone and kidney damage, leading to extreme pain—hence the name "itai-itai" or "it hurts, it hurts" disease.

The experiences (of these victims) ultimately served as a wake-up call for the nation. Public protests and legal battles forced the government to establish stricter environmental regulations and compensate the victims. The lessons learned from these tragedies led to the creation of the Environment Agency and laid the groundwork for Japan's modern environmental protection policies. They stand as a powerful reminder that true progress must be sustainable and that ignoring environmental consequences will always come at a price.

設問

問 78. What is the main idea of this passage?

- (a) The history of Japanese environmental law.
- (b) The tragic consequences of Minamata and Itai-itai diseases.
- (c) The devastating legacy of Japan's Four Major Pollution Diseases caused by rapid industrialization.

- (d) The importance of eating fish and rice as part of a healthy diet.

問 79. According to the passage, what was a contributing factor to the severity of Minamata disease's impact?

- (a) The company's swift response to the problem.
(b) The victims' immediate access to medical treatment.
(c) The consumption of contaminated fish by local residents.
(d) A lack of public awareness about the disease.

問 80. In the second paragraph, what is the closest meaning of the underlined word plight?

- (a) a joyful experience
(b) a difficult or dangerous situation
(c) a type of illness
(d) a legal battle

問 81. The passage suggests that the victims of these pollution diseases served as a "wake-up call" for Japan. What does this phrase mean in this context?

- (a) The government was forced to acknowledge the problem and take action.
(b) The victims were able to recover completely from their illnesses.
(c) The diseases helped to boost the economy.
(d) The public became less interested in environmental issues.

解答と解説

問 78

正解

(c) The devastating legacy of Japan's Four Major Pollution Diseases caused by rapid industrialization.

解説

この文章は、日本の高度経済成長がもたらした「暗い遺産」として、4つの公害病を全体的に説明しています。特に、それらが急速な工業化によって引き起こされたことを強調しています。

重要単語の解説:

legacy (遺産) : 過去から受け継がれたもの。良いものにも悪いものにも使われる。

unregulated (規制されていない) : 法律や規則による制限がないこと。

devastating (破壊的な、壊滅的な) : 非常に大きな損害を与えるさま。

問 79

正解

(c) The consumption of contaminated fish by local residents.

解説

第2段落に、「水俣病は、水銀に汚染された化学工場の廃水が水俣湾に排出され、その水銀が魚介類に蓄積し、それを食べた地元住民によって引き起こされた」と明確に書かれています。

重要単語の解説:

contaminated (汚染された) : 有害な物質などが混入していること。

accumulate (蓄積する) : 徐々に集まっていくこと。

neurological disorders (神経障害) : 脳、脊髄、神経に影響を及ぼす病気。

問 80

正解

(b) a difficult or dangerous situation

解説

****plight**** (名詞) は、「苦境、窮状」を意味します。ここでは、水俣病の被害者たちが直面した、困難で危険な状況を指しています。

重要単語の解説:

plight (苦境、窮状)

問 81

正解

(a) The government was forced to acknowledge the problem and take action.

subsubsection*解説最終段落に、「これらの被害者の経験は、最終的に国にとっての**警鐘** (wake-up call) となった」とあります。これは、公衆の抗議や法廷闘争が、政府に「**より厳しい環境規制を確立し、被害者を補償すること**」を強いたことを意味しています。vspace1em

重要単語の解説:

wake-up call (警鐘): 問題に気づかせ、行動を促すきっかけ。

stricter (より厳しい)**compensate** (補償する): 損害や苦痛に対して金銭などを支払うこと。

0.21 AI と倫理

次の英文を読み、後の設問に答えなさい。

The AI Trolley Problem: Who Should Self-Driving Cars Protect?

トロリー

As Artificial Intelligence (AI) becomes more integrated into our daily lives, it brings with it new ethical dilemmas. The most pressing of these might be the "AI trolley problem," a modern take on a classic philosophical thought experiment. The core of this problem is simple: if a self-driving car is about to face an unavoidable crash, what moral decisions should its algorithm be programmed to make?

Consider this scenario: a self-driving car is traveling down a street when a pedestrian suddenly jumps out in front of it. The car's only options are to either hit the pedestrian, or swerve into a wall, which would **endanger** the passenger inside the car. Which choice should the AI make? One ethical framework, utilitarianism, would argue that the car should make the choice that saves the greatest number of lives. However, this raises another question: should the life of a single pedestrian be valued more than the life of the car's owner?

Another framework, deontology, suggests that certain actions are inherently right or wrong, regardless of their consequences. A deontologist might argue that the car should never be programmed to intentionally harm its passenger, as doing so would violate a fundamental duty to protect them. This principle, however, might lead to the death of innocent pedestrians, which also feels morally unacceptable. As an alternative, should the AI be programmed to simply minimize damage to the car itself, to avoid legal liability and economic loss? This approach seems deeply flawed.

The debate over these issues is far from over. Unlike human drivers who react instinctively in a split second, an AI's decision must be pre-programmed, forcing us to explicitly define our moral values. There is no easy answer, and the way societies resolve these ethical challenges will not only determine the future of technology but also reflect our collective understanding of morality.

設問

問 82. What is the main topic of the passage?

- (a) The history of the trolley problem.
- (b) The legal issues of self-driving car manufacturers.
- (c) The ethical dilemmas facing AI in self-driving cars.
- (d) The technology behind self-driving cars.

- 問 83. According to the passage, what would a utilitarian approach suggest for the self-driving car?
- (a) It should always prioritize the passenger's life.
 - (b) It should minimize damage to the car.
 - ☒ (c) It should make the choice that saves the most lives.
 - (d) It should always protect pedestrians.
- 問 84. In the second paragraph, what is the closest meaning of the underlined word endanger?
- (a) to provide comfort to
 - (b) to put in a safe place
 - ☒ (c) to cause to be in a dangerous situation
 - (d) to make a difficult decision
- 問 85. What makes the AI trolley problem different from the human trolley problem, according to the passage?
- (a) AI can make decisions much faster than humans.
 - ☒ (b) AI's decisions must be pre-programmed with explicit moral values.
 - (c) AI is not affected by emotions.
 - (d) AI can communicate with passengers about the best choice.

解答と解説

問 82

正解

(c) The ethical dilemmas facing AI in self-driving cars.

解説

文章全体が、自動運転車が事故に直面した際に、どの命を優先すべきかという**倫理的なジレンマ** (ethical dilemmas) について論じています。

重要単語の解説:

integrated (統合された): 複数のものが一つにまとめられている状態。

dilemmas (ジレンマ): 二つの選択肢があり、どちらを選んでも望ましくない結果になる板挟みの状況。

ethical (倫理的な): 道徳や規範に関わるさま。

問 83

正解

(c) It should make the choice that saves the most lives.

解説

第2段落に、「**功利主義** (utilitarianism) という一つの倫理的枠組みは、車が**最も多くの命を救う選択**をすべきだと主張するだろう」と書かれています。

重要単語の解説:

scenario (シナリオ): 想定される状況や筋書き。

utilitarianism (功利主義): 最大多数の最大幸福を目指す倫理思想。

framework (枠組み): 思考や行動の土台となる構造。

問 84

正解

(c) to cause to be in a dangerous situation

解説

****endanger**** (動詞) は、「危険にさらす」という意味です。文脈では、車が壁に衝突することで、乗客が危険な状況に置かれることを指しています。

重要単語の解説:

endanger (危険にさらす)

問 85

正解

(b) AI' s decisions must be pre-programmed with explicit moral values.

subsubsection*解説最終段落に、「一瞬で本能的に反応する人間のドライバーと異なり、AI の決定は**事前にプログラムされなければならない**。そのため、私たちは**道徳的価値を明確に定義する**ことを余儀なくされる」と述べられています。vspace1em

重要単語の解説:

explicit (明白な、明確な)

instinctively (本能的に)

pre-programmed (事前にプログラムされた)

0.22 流行と集団心理

次の英文を読み、後の設問に答えなさい。

The Psychology of Trends: Why We Follow the Crowd

群衆

From fashion and music to viral social media challenges, trends are a constant force in modern society. We often see them as a simple matter of taste, but the desire to follow a trend is deeply rooted in our psychology. Understanding the social and psychological factors that drive these collective behaviors can help us see the world and ourselves in a new light.

One of the main reasons we follow trends is our innate need for social **conformity**. Humans are social creatures, and fitting in with a group often provides a sense of security and belonging. Adopting the same style of clothing or using the same slang as our peers can be a way of signaling our membership in a particular group. This desire to avoid being an outcast can be a powerful motivator. The trend cycle, in which new styles quickly replace old ones, also fuels this behavior as people rush to stay up-to-date.

Trends also play a crucial role in shaping our individual identity. While it may seem paradoxical, following a trend can be a way of expressing who we are. For example, by wearing a specific brand or listening to a certain genre of music, we are not just conforming to a group; we are also communicating our values, interests, and even our social status. A trend is a shared language that allows us to connect with others who are a part of our subculture. However, the line between expressing individuality and simply **adhering** to a trend can sometimes become blurred.

ぼやけた

In conclusion, trends are much more than passing fads. They are a reflection of our fundamental psychological needs for belonging and identity. As we navigate the constant flow of new trends, it is important to be aware of the underlying motivations. This awareness can help us to be more conscious consumers and to make choices that truly reflect our authentic selves, rather than just following the crowd.

根底にある

設問

問 86. What is the main idea of the passage?

- (a) The history of fashion and music trends.
- (b) The technical aspects of social media trends.
- (c) The psychological and social reasons people follow trends.
- (d) Why all trends are negative and should be avoided.

問 87. According to the second paragraph, what is one key reason people follow trends?

- (a) To express their unique individuality.
- (b) To be seen as an outcast.
- (c) To feel a sense of security and belonging.
- (d) To learn new languages and slang.

問 88. In the third paragraph, what is the closest meaning of the underlined word adhering?

- (a) to strongly support or stick to something
- (b) to ignore or reject something
- (c) to create or invent something new
- (d) to question something's value

問 89. What paradoxical idea is presented in the third paragraph?

- (a) That trends are both old and new at the same time.
- (b) That following a trend can be a way of expressing one's individuality.
- (c) That a trend can be popular and unpopular at the same time.
- (d) That trends are a powerful force but have no real impact.

解答と解説

問 86

正解

(c) The psychological and social reasons people follow trends.

解説

この文章は、第2段落で「社会的同調 (social conformity)」という心理的な要因を、第3段落で「個人のアイデンティティ」という社会的な側面を説明しています。このように、流行を追いかけることの**心理的・社会的背景**を主題としています。

重要単語の解説:

trend (流行、傾向) : 一時的に広まる社会現象。

rooted (根付いた) : 深く結びついていること。

collective behavior (集団行動) : 多くの人が似た行動をとること。

問 87

正解

(c) To feel a sense of security and belonging.

解説

第2段落に、「グループに溶け込むことは、しばしば**安心感と帰属意識** (a sense of security and belonging) をもたらす」と書かれています。流行を追うのは、仲間外れになるのを避けるためです。

重要単語の解説:

innate (生来の、生まれつきの) : もともと備わっている性質。

conformity (同調) : 周囲に合わせて行動すること。

outcast (のけ者、仲間外れ) : 集団から外された人。

問 88

subsubsection*正解 **(a) to strongly support or stick to something**

解説

adhere (動詞) は、「(主義や規則などに) 固く忠実である、固執する」という意味です。ここでは、単に流行に従うことを指しています。

重要単語の解説:

adhere (固執する、忠実に守る)

paradoxical (逆説的な) : 一見矛盾しているように見えるさま。

問 89

正解

(b) That following a trend can be a way of expressing one's individuality.

解説

第3段落の冒頭に、「流行を追うことは、**逆説的**に思えるかもしれないが、自分自身は何者であるかを表現する方法になり得る」と書かれています。これは、流行に従いながらも、その選択を通じて個性を表現するという矛盾した考え方です。vspace1em

重要単語の解説:

individuality (個性、独自性) : その人や物だけが持つ特徴。

subculture (サブカルチャー) : 主流文化とは異なる、特定の集団が共有する文化。**authentic** (本物の、本当の) : 偽りや見せかけではないさま。

0.23 日本のアニメの海外影響

次の英文を読み、後の設問に答えなさい。

Beyond the Screen: How Japanese Anime Shapes Global Culture

For decades, Japanese animation, or anime, has grown from a niche interest into a powerful global cultural force. It is no longer just a form of entertainment; it has become a significant ****vehicle**** for cultural exchange, influencing everything from fashion and language to art and social values. The journey of anime from Japan's domestic market to international **prominence** is a fascinating case study in globalization. プロミネンス

One of the most remarkable impacts of anime is its ability to build communities and foster a sense of belonging among fans worldwide. Through online forums, social media, and conventions like Anime Expo, people from diverse backgrounds can connect over their shared passion. Anime serves as a common ground, helping to bridge geographical and linguistic barriers. This global fandom has created a vibrant subculture that celebrates Japanese storytelling and visual artistry.

Anime's influence also extends into the creative industries. Its distinctive visual style—characterized by large, expressive eyes, dynamic action sequences, and unique character designs—has inspired countless artists, filmmakers, and video game designers around the world. Major Hollywood studios have adopted anime's storytelling techniques and **aesthetic**, and its influence can be seen in popular movies and television shows. 美的 Moreover, the economic impact is immense, with merchandise, licensing, and streaming services generating billions of dollars globally.

In conclusion, the influence of Japanese anime goes far beyond the television screen. It is a powerful cultural export that has not only captivated audiences but also transformed creative industries and built global communities. Its continued rise shows that a nation's cultural products, when unique and well-executed, can become a source of soft power that truly transcends borders.

設問

問 90. What is the main idea of the passage?

- (a) The history of Japanese anime production.
- (b) The unique visual characteristics of anime.
- (c) The financial success of the anime industry.
- ☒ (d) The widespread cultural and economic influence of anime around the world.

問 91. According to the second paragraph, how does anime help to build global communities?

- (a) It is broadcast in many different languages.
- ☒ (b) It helps people with diverse backgrounds connect over a shared interest.
- (c) It provides free merchandise to fans at conventions.
- (d) It encourages people to learn Japanese.

問 92. In the first paragraph, what is the closest meaning of the underlined word ***vehicle***?

- (a) a type of car or bus
- ☒ (b) a method or means of doing something
- (c) an obstacle or problem
- (d) a temporary trend

問 93. What is NOT mentioned in the passage as a way anime has influenced other creative industries?

- (a) Its distinctive visual style.
- (b) Its storytelling techniques.
- (c) Its aesthetic influence on Hollywood movies.
- ☒ (d) Its use of traditional Japanese music.

解答と解説

問 90

正解

(d) The widespread cultural and economic influence of anime around the world.

解説

文章全体が、日本のアニメが海外の文化や経済に与えた広範な影響（widespread influence）について論じています。文化交流、コミュニティ形成、他産業への影響、経済的な成功といった多角的な側面が説明されています。

重要単語の解説:

niche interest (ニッチな興味)：特定の少数の人々にしか知られていない興味。

prominence (卓越、著名)：広く知られていること、目立つこと。

globalization (グローバル化)：世界が一体となること。

問 91

正解

(b) It helps people with diverse backgrounds connect over a shared interest.

解説

第2段落に、「多様な背景を持つ人々が、共通の情熱を通じて**つながる**ことができる」と書かれています。アニメは共通の基盤（common ground）となり、地理的・言語的な障壁を越えるのに役立っています。

重要単語の解説:

foster (育む、促進する)

diverse backgrounds (多様な背景)：人種、文化、国籍などが異なること。

bridge (橋渡しをする)：隔たりを埋める。

問 92

正解

(b) a method or means of doing something

解説

****vehicle**** (名詞) は、文字通り「乗り物」を意味しますが、ここでは比喩的に、目的を達成するための「手段」や「媒体」を意味します。アニメが文化交流のための手段として機能していることを示しています。

重要単語の解説:

vehicle (手段、媒体)

問 93

正解

(d) Its use of traditional Japanese music.

解説

第3段落で、アニメが他の産業に影響を与えた点として、**「独特な視覚スタイル」「ストーリーテリングの技術」「ハリウッドへの美的影響」**が挙げられていますが、**「伝統的な日本の音楽の使用」**については言及されていません。

重要単語の解説:

distinctive (独特の、特徴的な)

aesthetic (美的、美学): 美の観点。

merchandise (商品、グッズ)

0.24 おまけと社会問題

次の英文を読み、後の設問に答えなさい。

The Unintended Consequences of Collectible Toys: A Case of Food Waste

In the late 1990s and early 2000s, a unique social phenomenon emerged in Japan, highlighting the complex relationship between consumerism and waste. This was the boom of candy toys, or ‘shokugan’, which gained immense popularity not for the candy itself, but for the highly detailed collectible toys inside. The most famous example was Furuta’s Choco Egg series, which featured intricately designed animal figures. However, this trend led to a serious and unintended consequence: the large-scale disposal of uneaten chocolate.

The primary goal of many consumers was to collect the entire series of miniature figures, some of which were rare secret items. To find these rare toys, people would buy boxes of Choco Eggs, often without any intention of consuming the chocolate. As a result, news reports and social media discussions showed piles of discarded chocolate and packaging near stores and in garbage bins. This created a new kind of social issue: ‘omake-gai’ (purchasing for the bonus item), which caused **significant** food waste.

This phenomenon forced both the public and companies to reconsider the ethical implications of their business models and consumption habits. While companies benefited from high sales, they faced public criticism for contributing to food waste. Consumers, on the other hand, had to confront the morality of their actions. The problem was not the toys or the chocolate themselves, but the system that encouraged a wasteful mindset. It served as a clear example of how a simple marketing strategy could have far-reaching and negative effects on society and the environment.

In conclusion, the Choco Egg boom was more than just a passing fad. It highlighted a fundamental problem in modern consumer culture, where the value of a product becomes secondary to its bonus item. This issue remains relevant today, as new collectibles and lottery-style marketing strategies continue to be used. It challenges us to be more conscious of our purchases and to consider the broader impact of our consumption.

設問

問 94. What is the main idea of the passage?

- (a) The history of Japanese ‘shokugan’ toys.
- (b) The popularity of Choco Eggs in Japan.

- ☒ (c) How a marketing trend led to a social problem of food waste.
- (d) The business strategies of Furuta company.

問 95. According to the second paragraph, what was the primary reason people bought Choco Eggs?

- (a) To enjoy the taste of the chocolate.
- ☒ (b) To collect the detailed animal figures inside.
- (c) To sell the chocolate at a higher price.
- (d) To trade the chocolate with friends.

問 96. In the second paragraph, what is the closest meaning of the underlined word ***significant***?

- (a) small and unimportant
- (b) not visible to the public
- ☒ (c) large and important
- (d) temporary and short-lived

問 97. What is the main problem identified in the third paragraph?

- (a) The toys were not well-designed.
- ☒ (b) The business model encouraged a wasteful mindset.
- (c) The chocolate was not delicious.
- (d) Companies received too much praise from the public.

解答と解説

問 94

正解

(c) How a marketing trend led to a social problem of food waste.

解説

文章は、チョコエッグという**マーケティング戦略** (marketing trend) が、大量の食品廃棄という**社会問題** (social problem) をどのように引き起こしたかを具体的に説明しています。

重要単語の解説:

phenomenon (現象): 注目すべき出来事や事象。

consumerism (消費主義): 物質的な消費を重んじる考え方。

unintended consequence (意図せぬ結果): 予測していなかった悪い結果。

問 95

正解

(b) To collect the detailed animal figures inside.

解説

第2段落の冒頭に、「多くの消費者の**主な目的** (primary goal) は、ミニチュアのフィギュアをコンプリートすることだった」と明確に書かれています。

重要単語の解説:

primary goal (第一の目的)

miniature (ミニチュア): 小さな模型や人形。

discarded (捨てられた)

問 96

正解

(c) large and important

解説

****significant**** (形容詞) は「重要な、意味のある」という意味です。ここでは「**大規模な**食品廃棄」という文脈で使われており、その深刻さや規模の大きさを強調しています。vspace1em

重要単語の解説:

significant (重要な、かなりの)

問 97

正解

(b) The business model encouraged a wasteful mindset.

解説

第3段落に、「問題は、おもちゃやチョコレートそのものではなく、**浪費する考え方を助長したシステム**にあった」と結論付けられています。マーケティング戦略が、消費者に無駄遣いを促したことが根本原因だと指摘しています。vspace1em

重要単語の解説:

ethical implications (倫理的影響): 倫理的な観点から生じる結果。

confront (直面する、立ち向かう)

mindset (考え方、物の見方)

0.25 クリントンとブッシュの移行期

次の英文を読み、後の設問に答えなさい。

The Prank War in the White House: The 2001 Presidential Transition

The 2001 presidential transition from Bill Clinton to George W. Bush was one of the most **contentious** in modern U.S. history, not just for the legal and political battles, but for the chaotic events that **unfolded behind closed doors**. After a highly disputed election, the incoming Bush team found their new White House offices had been subjected to a series of **pranks** and acts of **vandalism**. What the Bush team experienced went far beyond simple **partisanship** and became a humorous, yet revealing, example of political rivalry.

閉ざされた扉の向こうで繰り広げられた
↓
非公開の場で展開される

Reports from Bush's staff detailed numerous instances of intentional **mischief**. The most widely reported incidents involved computer keyboards, with staff discovering that many of the ****W**** keys had been removed or swapped with other letters. Phone lines were disconnected, drawers in desks were glued shut, and even office furniture was found to be **vandalized**. More disturbingly, some offices had **obscene** messages and drawings scrawled on the walls and pictures of the incoming president were **defaced**. The cost of the damage was estimated to be in the tens of thousands of dollars.

The response from the Clinton administration was mixed. While some officials dismissed the incidents as the harmless actions of a few **rogue** individuals, the Bush team characterized them as a serious breach of professional **decorum** and a violation of the tradition of a peaceful transfer of power. The White House Chief of Staff at the time, Andrew Card, publicly expressed his frustration, stating that he was "disappointed" by the lack of respect shown to his new staff.

In conclusion, the 2001 transition stands as a unique and memorable moment in U.S. political history. While these acts were not a threat to national security, they revealed the deep personal **animosity** that existed between the two political parties. The pranks, whether childish or malicious, serve as a reminder that behind the formal titles and grand ceremonies, politics is often driven by human emotions—rivalry, frustration, and a desire to get the last laugh.

設問

問 98. What is the main idea of the passage?

- (a) The political and legal battles of the 2000 U.S. election.
- (b) The serious damage to White House property during the 2001 transition.
- (c) That the transition from Clinton to Bush involved significant pranks and reflected deep political

rivalry.

- (d) The cost of repairing the White House after the transition.

問 99. According to the second paragraph, which of the following was NOT an incident reported by the Bush team?

- (a) Disconnected phone lines.
(b) The removal of "W" keys from keyboards.
(c) The presence of obscene drawings on walls.
(d) Damaged computer monitors.

問 100. In the first paragraph, what is the meaning of the word ***"contentious"***?

- (a) calm and peaceful
(b) highly debated and disputed
(c) humorous and friendly
(d) boring and uneventful

問 101. What did the Bush team consider the pranks to be?

- (a) A harmless and humorous gesture.
(b) A serious breach of professional behavior.
(c) A tradition that should be continued.
(d) An act of national security threat.

解答と解説

問 98

正解

(c) That the transition from Clinton to Bush involved significant pranks and reflected deep political rivalry.

解説

文章全体を通して、クリントンからブッシュへの政権移行が、**多くの嫌がらせ** (pranks) を含み、それが両陣営間の**根深い政治的対立** (deep political rivalry) を反映していたことを説明しています。

重要単語の解説:

transition (移行): ある政権から別の政権へ移ること。

prank (いたずら、悪ふざけ)

rivalry (対立、競争): 敵意を持って競い合うこと。

問 99

正解

(d) Damaged computer monitors.

解説

第2段落には、「**W**キーが外されていたこと」、「電話線が切断されていたこと」、「卑猥な落書き」などが報告されたと書かれています。しかし、「**コンピューターのモニターが壊されていた**」という記述はありません。

重要単語の解説:

intentional mischief (意図的ないたずら)

vandalism (器物損壊): 公共物や他人の物を意図的に壊す行為。

obscene (わいせつな、不快な)

scrawled (乱雑に書かれた)

問 100

正解

(b) highly debated and disputed

解説

****contentious**** (形容詞) は、「議論的となる、争いを引き起こす」という意味です。本文の最初の文では、「**法的な、そして政治的な戦い** (legal and political battles)」という言葉と組み合わせて使われており、激しい対立があったことを示唆しています。vspace1em

重要単語の解説:**contentious** (議論の的となる)**unfolded** (展開された、明らかになった)**問 101****正解**

(b) A serious breach of professional behavior. subsection*解説第 3 段落に、「ブッシュ陣営はそれらの行為を、プロフェッショナルな礼儀作法に対する**深刻な違反** (a serious breach of professional decorum) だと見なした」と書かれています。クリントン側が「無害な行為」と捉えたのに対し、ブッシュ側はより厳しく非難しました。vspace1em

重要単語の解説:**breach** (違反、侵害)**decorum** (礼儀正しさ、作法)**manifest** (明らかにする、表に出す)

0.26 世界の原発事故

次の英文を読み、後の設問に答えなさい。

Three Mile Island, Chernobyl, and Fukushima: A Comparative Analysis of Nuclear Disasters

The history of nuclear energy is marked by three major disasters that profoundly shaped public perception and policy: Three Mile Island in the United States, Chernobyl in the Soviet Union, and Fukushima in Japan. While each event involved a nuclear power plant, their causes, consequences, and global reactions were distinct, offering critical lessons in safety, human error, and environmental vulnerability.

The Three Mile Island accident in 1979 was the least severe of the three. It was caused by a mechanical failure and a combination of operator errors, which led to a partial meltdown of the reactor core. Crucially, the reactor's containment building held, preventing a significant release of radioactive material into the atmosphere. The accident led to widespread public concern and a major reform of nuclear safety regulations in the U.S. and around the world, but it resulted in no direct deaths from radiation exposure.

In stark contrast, the Chernobyl disaster in 1986 was a catastrophic event. A flawed reactor design, coupled with poorly trained operators who disregarded safety protocols during a test, caused a massive power surge. This resulted in a steam explosion and a graphite fire that blew the reactor's roof off, releasing a huge plume of radioactive fallout into the atmosphere. The Soviet government's slow and secretive response worsened the tragedy, and the long-term health consequences from radiation poisoning were severe for both the local population and emergency workers.

The Fukushima disaster in 2011 was unique in that it was triggered by a natural disaster. A powerful earthquake and subsequent tsunami overwhelmed the plant's defenses, leading to the failure of the cooling systems. This caused a series of meltdowns and hydrogen explosions, which released radioactive material. While the direct cause was external, the accident exposed vulnerabilities in the plant's design and emergency preparedness. It led to a re-evaluation of nuclear energy safety standards worldwide, particularly regarding natural disaster readiness.

In conclusion, these three events highlight different facets of nuclear risk. Three Mile Island was a lesson in human and mechanical failure, leading to a focus on operational safety. Chernobyl was a grim reminder of the dangers of poor design and a lack of transparency. Fukushima demonstrated the vulnerability of even modern systems to natural forces. By studying these distinct disasters, we can better understand the ongoing challenges and responsibilities associated with nuclear power.

設問

問 102. What is the main idea of the passage?

- (a) Nuclear power plants are extremely dangerous and should all be shut down.
- (b) The Three Mile Island accident was the most serious nuclear disaster in history.
- (c) The three major nuclear accidents had distinct causes and consequences.
- (d) Human error was the primary cause of all three nuclear disasters.

問 103. According to the passage, what was a key factor in the Three Mile Island accident's relatively low severity?

- (a) The operators were highly trained and corrected the error quickly.
- (b) The accident was caused by a natural disaster.
- (c) The reactor's containment building successfully held.
- (d) There was no mechanical failure involved.

問 104. In the fourth paragraph, what is the meaning of the word *****"vulnerabilities"*****?

- (a) strengths or advantages
- (b) hidden dangers or weaknesses
- (c) new technologies or designs
- (d) public fears or concerns

問 105. What does the passage identify as a key lesson from the Chernobyl disaster?

- (a) The need for better cooling systems.
- (b) The dangers of poor reactor design and a lack of transparency.
- (c) The importance of natural disaster preparedness.
- (d) The unimportance of operator training.

解答と解説

問 102

正解

(c) The three major nuclear accidents had distinct causes and consequences.

解説

文章は、スリーマイル島、チェルノブイリ、福島 の 3 つの事故が、それぞれ「原因 (causes)」と「結果 (consequences)」において**異なっていた** (distinct) ことを比較分析しています。これが本文全体の主題です。

重要単語の解説:

distinct (明確に異なる、別個の)

profoundly (深く、大いに): 影響や変化の度合いが非常に大きいこと。

perception (認識、見方)

問 103

正解

(c) The reactor's containment building successfully held. subsection*解説第 2 段落に、「決定的に重要なこととして、原子炉の**格納容器が持ちこたえ** (the reactor's containment building held)、大量の放射性物質が大気中に放出されるのを防いだ」と書かれています。これが、この事故の深刻度が低かった主な要因です。 vspace1em

重要単語の解説:

crucially (決定的に、極めて重要に)

containment building (格納容器): 原子炉を覆い、放射性物質の漏洩を防ぐための建物。

問 104

subsection*正解 **(b) hidden dangers or weaknesses** subsection*解説**vulnerabilities** (名詞) は「脆弱性、弱点」という意味です。本文では、福島第一原発の事故が、**設計上の弱点**や**緊急時の準備不足** (vulnerabilities in the plant's design and emergency preparedness) を露呈したと述べています。 vspace1em

重要単語の解説:

vulnerability (脆弱性、弱点)

overwhelm (圧倒する、制圧する)

preparedness (準備、備え)

問 105

正解

(b) The dangers of poor reactor design and a lack of transparency. subsubsection*解説最終段落に、「チェルノブイリは、**ずさんな原子炉設計**と**透明性の欠如**の危険性についての
厳しい教訓であった」と明記されています。 vspace1em

重要単語の解説:

grim (厳しい、暗い)：希望がなく、悲惨な様子。

transparency (透明性)：情報が隠されずに公開されていること。

disregarded (無視した、軽視した)

0.27 ベトナム戦争とペンタゴン・ペーパーズ

次の英文を読み、後の設問に答えなさい。

The Power of a Pen: How the Pentagon Papers Exposed Government Deceit and Changed American Society

The Vietnam War was one of the most divisive conflicts in U.S. history, fueling widespread public protest and a deep mistrust of the government. At the heart of this turmoil was the public's growing suspicion that they were not being told the full truth about the war. This suspicion was confirmed in 1971 by the publication of the ****Pentagon Papers****, a classified document that revealed the government's secret history of the conflict. This event did not just expose facts; it fundamentally altered the relationship between the American people, their government, and the press.

The Pentagon Papers were a top-secret study commissioned by the Department of Defense. It detailed the U.S. political and military involvement in Vietnam from the 1940s to 1968. The report was leaked to the press by Daniel Ellsberg, a military analyst who had worked on the study and had become disillusioned with the war. He believed the American public had a right to know the truth. The documents revealed that successive U.S. administrations had systematically misled the public about the true intentions and progress of the war, escalating the conflict while assuring the public that they were seeking peace.

The New York Times was the first newspaper to publish excerpts from the classified report. The Nixon administration immediately sought a court injunction to stop the publication, citing a threat to national security. However, in a landmark decision, the Supreme Court ruled in favor of the newspapers, upholding the freedom of the press and the public's right to information. This legal battle became a symbol of the struggle between government secrecy and the press's role as a public watchdog.

The impact of the Pentagon Papers was immense. For many Americans, it confirmed their worst fears: that their government had been dishonest. It deepened the existing anti-war sentiment and fueled a more skeptical attitude towards official statements. More importantly, it established a powerful precedent for investigative journalism, showing that a free press could act as a crucial check on government power. The event forever changed how the American public viewed its leaders and how journalists approached their role in society.

設問

問 106. What is the main idea of the passage?

- (a) The Pentagon Papers were a series of documents written by Daniel Ellsberg.
- (b) The Vietnam War led to a major change in U.S. foreign policy.
- (c) The Pentagon Papers exposed government deception and had a lasting impact on American society and journalism.
- (d) The Supreme Court always rules in favor of the freedom of the press.

問 107. According to the second paragraph, what was revealed in the Pentagon Papers?

- (a) The government's plan to end the war immediately.
- (b) That the government had been honest about the war's progress.
- (c) The government's true intentions and a systematic effort to mislead the public.
- (d) A new technology for fighting the war.

問 108. In the first paragraph, what is the closest meaning of the word **“divisive”**?

- (a) uniting people
- (b) causing strong disagreement
- (c) a quick and short conflict
- (d) a clear victory

問 109. What was the outcome of the Supreme Court's ruling on the publication of the Pentagon Papers?

- (a) It forced the newspapers to stop publishing the documents.
- (b) It upheld the government's right to secrecy.
- (c) It ruled against the newspapers, but the public supported them.
- (d) It upheld the freedom of the press and the public's right to information.

解答と解説

問 106

正解

(c) The Pentagon Papers exposed government deception and had a lasting impact on American society and journalism.

解説

文章は、ペンタゴン・ペーパーズが政府の**欺瞞**を暴露したこと、そしてそれがアメリカ社会とジャーナリズムに**永続的な影響**を与えたことを中心に論じています。

重要単語の解説:

deception (欺瞞、詐欺)：人を欺く行為。

lasting (永続的な)：長く続く、恒久的な。

fundamentally (根本的に)：根底から、抜本的に。

問 107

subsubsection*正解 **(c) The government's true intentions and a systematic effort to mislead the public.** subsubsection*解説第2段落に、「文書は、歴代の米政権が、戦争の真の意図と進捗について国民を**組織的に欺いてきた** (systematically misled the public) ことを明らかにした」と書かれています。vspace1em

重要単語の解説:

commissioned (依頼された、委託された)

disillusioned with (～に幻滅した)

escalating (エスカレートさせていた、拡大させていた)

問 108

subsubsection*正解 **(b) causing strong disagreement** subsubsection*解説**divisive** (形容詞)は「意見の対立を引き起こす、分裂を招く」という意味です。本文では、「**広範な大衆の抗議** (widespread public protest)」や「**深い不信感** (deep mistrust)」といった言葉と組み合わせて使われており、国民の間で強い意見の対立があったことを示しています。vspace1em

重要単語の解説:

divisive (分裂的な)

turmoil (騒乱、混乱)：社会的な動揺や無秩序な状態。

問 109

subsubsection*正解 **(d) It upheld the freedom of the press and the public's right to information.** subsubsection*解説第3段落に、「最高裁は、報道の自由と国民が情報を得る権利を**

支持** (upheld) し、新聞社側に有利な画期的な判決を下した」と明確に述べられています。vspace1em

重要単語の解説:

injunction (差し止め命令)：裁判所が出す、特定の行為を禁じる命令。

landmark (画期的な)：歴史に残るほど重要な出来事。

upholding (支持する、維持する)：物事の正当性や原則を守ること。

0.28 パディントンという名のクマ

次の英文を読み、後の設問に答えなさい。

More Than Just a Bear: The Enduring Legacy of Paddington

For generations of children and adults around the world, Paddington Bear is a beloved figure—a small, polite bear from Peru who arrives at Paddington Station in London with a suitcase, a half-eaten jar of marmalade, and a label around his neck that reads, "Please look after this bear. Thank you." Written by Michael Bond, the first book, *A Bear Called Paddington*, was published in 1958. However, Paddington's story is far more than a simple tale of a cute bear; it is a profound commentary on themes of immigration, identity, and the importance of kindness.

Paddington's journey from "Darkest Peru" to a bustling London station can be seen as a metaphor for the experience of a displaced person or a refugee. He arrives in a strange country, alone and with little but the clothes on his back. The Brown family, who discover him, do not turn him away. Instead, they offer him a home, a new name, and a place within their family. The narrative demonstrates how compassion and acceptance can transform the life of an outsider and enrich the lives of those who welcome him. The story challenges the idea that "foreigners" are to be feared, instead portraying them as individuals who can contribute to a new society.

The character of Paddington himself embodies a unique blend of a British gentleman and a Peruvian outsider. He is consistently polite, well-mannered, and fond of tea—traits often associated with British culture. At the same time, his love for marmalade and his occasional clumsy adventures remind us of his Peruvian roots. This dual identity serves to make him a relatable character for anyone who has ever felt like an outsider. It suggests that a person's identity is not a single, fixed thing, but a fluid mix of their background and their new environment.

Ultimately, Paddington's lasting appeal lies in his simple yet powerful message. Through his humorous misadventures, the stories teach us about tolerance and empathy. The world of Paddington is one where kindness is always rewarded, and differences are celebrated rather than scorned. In an increasingly complex and divided world, the image of a small bear finding a home and a family offers a hopeful reminder that with a little bit of kindness, we can all find a place to belong.

設問

問 110. What is the main idea of the passage?

- (a) Paddington Bear is the most popular children's book character in the world.

- (b) The story of Paddington is a metaphor for immigration and acceptance.
- (c) The story of Paddington is about a bear's search for his parents.
- (d) Michael Bond wrote the book based on his own experiences in Peru.

問 111. According to the second paragraph, what does the Brown family's action towards Paddington represent?

- (a) The importance of following strict rules and traditions.
- (b) The fear of people from different countries.
- (c) The power of compassion and acceptance towards outsiders.
- (d) The idea that all new arrivals should be named after their destination.

問 112. In the fourth paragraph, what is the meaning of the word **scorned**?

- (a) appreciated or praised
- (b) understood or learned
- (c) disrespected or mocked
- (d) studied or analyzed

問 113. How is Paddington's identity described in the third paragraph?

- (a) As being exclusively British.
- (b) As a single, fixed identity.
- (c) As a mix of his Peruvian background and British culture.
- (d) As completely forgetting his Peruvian roots.

解答と解説

問 110

正解

(b) The story of Paddington is a metaphor for immigration and acceptance.

subsubsection*解説文章全体が、パディントンの物語を単なる児童文学ではなく、「移民の経験の**比喻** (metaphor for the experience of a displaced person)」として捉え、**受容** (acceptance) と**優しさ**の重要性を論じています。

重要単語の解説:

enduring (永続的な): 長く続く、不朽の。

profound (深遠な): 表面的なものではなく、深い意味や影響を持つ。

commentary (論評、解説): 特定の主題についての意見や考察。

問 111

subsubsection*正解 **(c) The power of compassion and acceptance towards outsiders.** subsubsection*解説第2段落に、「彼(パディントン)を発見したブラウン一家は彼を追い返さなかった。代わりに、家と居場所を与えた。この物語は、**思いやり** (compassion) と**受容** (acceptance) が部外者の人生をどう変えるかを示している」と明確に述べられています。vspace1em

重要単語の解説:

displaced person (避難民、難民): 戦争や災害などで故郷を追われた人々。

metaphor (比喻、暗喩): ある事柄を別の事柄になぞらえて表現する修辞法。

compassion (思いやり、同情)

問 112

subsubsection*正解 **(c) disrespected or mocked** subsubsection*解説**scorned** (動詞 scorn の過去形) は、「軽蔑する、あざ笑う」という意味です。第4段落では、「違いは**軽蔑される** (scorned) のではなく、**称賛される** (celebrated) べきだ」と述べられており、対義語の「celebrated」から意味を推測することができます。vspace1em

重要単語の解説:

tolerance (寛容): 自分と異なる考えや行動を受け入れること。

empathy (共感): 他者の感情や経験を理解し、共有すること。 **scorn** (軽蔑する、あざ笑う)

問 113

subsubsection*正解 **(c) As a mix of his Peruvian background and British culture.** subsubsection*解説第3段落に、「パディントンのキャラクターは、**イギリス紳士**と**ペルーのよそ者**という独自の組み合わせを体現している」と書かれています。紅茶好きで礼儀正しい英国的な側面と、マーマレード好きで不器用なペルーのルーツが混ざり合っていると説明されています。vspace1em

重要単語の解説:

embodied (体現する) : 思想や性質などを具体的な形で表すこと。

dual identity (二重のアイデンティティ)

fluid (流動的な、固定されていない)

0.29 彗星の謎

次の英文を読み、後の設問に答えなさい。

Cosmic Travelers: Unveiling the Secrets of Comets

For centuries, comets have captured the human imagination. Their sudden appearance in the night sky, often with a brilliant tail, was once seen as a mysterious omen, a sign of impending change or disaster. Today, thanks to modern astronomy, we understand that these celestial visitors are not mystical harbingers but are, in fact, "dirty snowballs" — a fascinating mixture of ice, dust, and rock orbiting the sun.

A comet is composed of three main parts: the nucleus, the coma, and the tail. The **nucleus**, the solid core, is typically a few kilometers wide and contains the majority of the comet's mass. As a comet approaches the sun, solar radiation heats the nucleus, causing the ice to sublimate directly into a gas. This creates a vast, temporary atmosphere around the nucleus called the **coma**. This gas and dust are then pushed away from the sun by a combination of solar wind and radiation pressure, forming the comet's characteristic **tail**. It is important to note that a comet's tail always points away from the sun, regardless of the comet's direction of travel.

Comets travel on highly elliptical orbits, which means they spend most of their time in the outer, colder regions of the solar system, far from the sun. The source of many comets is thought to be the **Oort Cloud**, a theoretical spherical shell of icy objects that surrounds our solar system. When the gravitational pull of a passing star or another celestial body disturbs an object in this cloud, it can be sent on a long, looping journey toward the inner solar system, where it becomes a visible comet.

While they may seem like mere spectacles, comets hold a wealth of scientific information. They are believed to be pristine relics from the formation of our solar system, carrying clues about the chemical composition of the early solar nebula. Some scientists even hypothesize that comets may have delivered water and organic molecules to early Earth, providing the building blocks for life. Thus, by studying these cosmic travelers, we are not only exploring the vastness of space but also piecing together the origins of our own planet.

設問

問 114. What is the main purpose of the passage?

- (a) To describe the history of comet observations.
- (b) To explain the components and origins of comets and their scientific importance.

- (c) To argue that comets are dangerous to the Earth.
- (d) To discuss the myths and legends associated with comets.

問 115. According to the second paragraph, what causes a comet's tail to form?

- (a) The comet's gravitational pull on a nearby planet.
- (b) The sunlight and solar wind pushing gas and dust away from the sun.
- (c) The comet's journey through a dense asteroid belt.
- (d) The sublimation of rock into ice.

問 116. In the second paragraph, what is the meaning of the word *******"sublimate"*******?

- (a) to melt from a solid to a liquid
- (b) to freeze from a liquid to a solid
- (c) to turn directly from a solid into a gas
- (d) to turn from a gas into a solid

問 117. What is the Oort Cloud believed to be?

- (a) A large group of asteroids within the inner solar system.
- (b) A gas giant planet in the outer solar system.
- (c) The main source of comets in our solar system.
- (d) The center of the galaxy.

解答と解説

問 114

正解

(b) To explain the components and origins of comets and their scientific importance. subsection*解説文章全体が、彗星の**構成要素** (nucleus, coma, tail)、その**起源** (Oort Cloud)、そして科学的な**重要性** (pristine relics, building blocks for life) について解説しています。

重要単語の解説:

unveiling (～の秘密を明らかにする): 隠されていたものや謎を解き明かすこと。

celestial (天体の、空の)

harbinger (先触れ、前兆): 何かが起こる兆候となるもの。

問 115

subsection*正解 **(b) The sunlight and solar wind pushing gas and dust away from the sun.** subsection*解説第 2 段落に、「このガスと塵は、**太陽風** (solar wind) と**放射圧** (radiation pressure) の組み合わせによって太陽から押しやられ、彗星の特徴的な尾を形成する」と書かれています。 vspace1em

重要単語の解説:

solar radiation (太陽放射): 太陽から放出されるエネルギー。

solar wind (太陽風): 太陽から放出される荷電粒子の流れ。

radiation pressure (放射圧): 光子が物体に与える圧力。

問 116

subsection*正解 **(c) to turn directly from a solid into a gas** subsection*解説**sublimate** (動詞) は、固体が液体を経由せず直接気体に変化する「昇華」を意味します。本文では、「太陽の熱によって、氷が直接ガスに昇華する (sublimate directly into a gas)」と説明されています。

重要単語の解説:

sublimate (昇華する): 固体が気体になる物理変化。

coma (コマ): 彗星の核の周りにあるガス状の大気。

characteristic (特徴的な)

問 117

subsection*正解 **(c) The main source of comets in our solar system.** subsection*解説第 3 段落に、「多くの彗星の起源は**オールトの雲** (Oort Cloud) であると考えられている」と書かれており、「太陽系を取り囲む、氷の天体からなる理論上の球状の殻」と説明されています。 vspace1em

重要単語の解説:

elliptical (楕円形の): 天体の軌道を表す言葉。

spherical (球状の)**Oort Cloud** (オールトの雲): 太陽系の最も外側にある、彗星の故郷とされる仮説上の領域。

0.30 量子力学と半導体

次の英文を読み、後の設問に答えなさい。

The Quantum Leap: From Abstract Theory to the Digital Age

When we hear the term "quantum mechanics," we often think of an abstract, confusing field of physics involving parallel universes and Schrödinger's cat. While it certainly explores strange and counterintuitive phenomena, quantum mechanics is not just a theoretical curiosity. In fact, it is the fundamental principle behind many of the technologies we use every day, most notably semiconductors, which are the building blocks of all modern electronics.

To understand the connection, we must first look at how materials behave. In classical physics, electrons in an atom are like planets orbiting a sun. However, quantum mechanics reveals that electrons exist in specific energy levels or "bands." Conductors, like copper, have electrons that are free to move, allowing electricity to flow easily. Insulators, like rubber, have electrons held tightly in a band, unable to move. Semiconductors, like silicon, are unique because they have a small energy gap between these bands. Under normal conditions, they act as insulators, but with a small amount of energy—from heat or light—electrons can jump across this gap and become mobile, allowing them to conduct electricity.

This phenomenon, known as the **quantum tunneling** effect, allows us to control the flow of electricity with extreme precision. In a transistor—the basic switch in a computer chip—a small voltage can be used to open or close a gate, allowing or blocking the flow of electrons. Without the principles of quantum mechanics governing how electrons behave in these energy bands, the design of transistors would be impossible. The entire digital world of ones and zeros is built upon this ability to precisely switch an electric current on and off.

In conclusion, the abstract world of quantum mechanics is not separate from our reality. It is the very foundation of the digital revolution. From the simple act of turning on a light to the complex operations of a supercomputer, the technology that shapes our lives is a direct application of quantum principles. The strange rules of the quantum world have been harnessed to create a future that classical physics could never have predicted.

設問

問 118. What is the main idea of the passage?

- (a) Quantum mechanics is an old and outdated theory of physics.
- (b) Conductors and insulators are the main components of computers.
- (c) The abstract principles of quantum mechanics are essential for modern semiconductor tech-

nology.

- (d) Transistors were invented before quantum mechanics was understood.

問 119. According to the second paragraph, what distinguishes a semiconductor from an insulator?

- (a) Semiconductors have no energy gap, allowing electrons to move freely.
- (b) Insulators have electrons that can easily jump across a small energy gap.
- (c) Insulators have a small energy gap, while semiconductors have a large one.
- (d) Semiconductors have a small energy gap, allowing electrons to jump across it under certain conditions.

問 120. In the third paragraph, what is the meaning of the word **“harnessed”**?

- (a) ignored or forgotten
- (b) misused or abused
- (c) utilized or controlled
- (d) discovered or found

問 121. How are the “ones and zeros” of the digital world related to quantum mechanics, according to the passage?

- (a) They are part of a philosophical concept called the “digital universe.”
- (b) They represent the on and off states of a transistor, which is controlled by quantum principles.
- (c) They are symbols used to represent different atoms in an electronic device.
- (d) They are related to the theory of parallel universes.

解答と解説

問 118

正解

(c) **The abstract principles of quantum mechanics are essential for modern semiconductor technology.** subsection*解説文章は、一見抽象的に見える量子力学の原理が、半導体技術という私たちの生活に不可欠な技術の**基礎**となっていることを中心に論じています。

重要単語の解説:

abstract (抽象的な): 具体性や実体がなく、観念的な。

counterintuitive (直感に反する): 直感的に理解しにくい、意外な。

fundamental (基礎的な、根本的な)

問 119

subsection*正解 (d) **Semiconductors have a small energy gap, allowing electrons to jump across it under certain conditions.** subsection*解説第2段落に、「半導体 (semiconductors) は、これらのバンドの間に**小さなエネルギーギャップ** (small energy gap) を持つ点でユニークである」と述べられており、熱や光といったエネルギーによって電子がこのギャップを飛び越えることができると説明されています。

重要単語の解説:

conductor (導体): 電気を通しやすい物質。

insulator (絶縁体): 電気を通しにくい物質。

energy gap (エネルギーギャップ): 物質中の電子が存在できないエネルギーの範囲。

問 120

subsection*正解 (c) **utilized or controlled** subsection*解説**harnessed** (動詞 harness の過去形) は、もともと馬に馬具を付けて制御する意味から、「(自然の力などを) **利用する、制御する**」という意味で使われます。ここでは、「量子世界の奇妙なルールが、未来を創るために**利用されてきた**」という意味になります。vspace1em

重要単語の解説:

harnessed (利用された、制御された)

revolution (革命): ある分野における根本的な変化。

supercomputer (スーパーコンピュータ)

問 121

subsection*正解 (b) **They represent the on and off states of a transistor, which is controlled by quantum principles.** subsection*解説第3段落に、「デジタル世界の**1

0** (ones and zeros) は、電流を正確にオン・オフするこの能力の上に築かれている」と述べられています。そして、そのスイッチングの能力は量子力学の原理によって可能になっていると説明されています。vspace1em

重要単語の解説:

transistor (トランジスタ): 電気信号を増幅またはスイッチングする半導体素子。

precision (精密さ、正確さ)

off and on (オンとオフ)

0.31 相対性理論と GPS

次の英文を読み、後の設問に答えなさい。

How Einstein's Theories Make Your GPS Work

When you use your phone to navigate, you are relying on a network of satellites orbiting the Earth, known as the Global Positioning System (GPS). These satellites are equipped with highly accurate atomic clocks that constantly broadcast their location and the time. Your phone's receiver then picks up these signals from multiple satellites to calculate your precise position. However, if it weren't for a specific part of Albert Einstein's work, your GPS would be completely useless.

The key concept is **time dilation**, a phenomenon predicted by Einstein's theory of relativity. The first part, the **Special Theory of Relativity**, states that time moves more slowly for an object the faster it travels. GPS satellites, traveling at speeds of about 14,000 kilometers per hour, experience a slight slowing of time relative to clocks on Earth. This effect is minuscule on its own, but over a single day, it amounts to a time difference of about 7 microseconds. While this may seem like a trivial amount, at the speed of light, this time difference would lead to a positioning error of over two kilometers per day.

The second part, the **General Theory of Relativity**, adds another layer of complexity. This theory states that gravity can also affect the passage of time. The stronger the gravitational field, the slower time passes. Because GPS satellites are orbiting high above the Earth, they experience a weaker gravitational pull than we do on the surface. As a result, their clocks actually run slightly faster—by about 45 microseconds per day—than clocks on the ground.

So, what is the net effect? The two relativistic effects work against each other, but the gravitational effect (General Relativity) is much stronger than the velocity effect (Special Relativity). Without any corrections, the satellites' clocks would run a total of about 38 microseconds faster per day than ground-based clocks. GPS engineers must precisely account for these time differences by adjusting the clocks on the satellites. By applying Einstein's theories, they ensure that the timing signals are accurate, allowing your GPS to pinpoint your location with remarkable accuracy, a direct testament to the practical power of theoretical physics.

設問

問 122. What is the main idea of the passage?

- (a) GPS satellites are equipped with highly accurate clocks.

- (b) GPS would be useless without accounting for the effects of Einstein's theories of relativity.
- (c) The General Theory of Relativity has no effect on GPS.
- (d) GPS satellites travel at very high speeds.

問 123. According to the passage, what effect does Special Relativity have on the clocks in GPS satellites?

- (a) It causes them to run faster than clocks on Earth.
- (b) It causes them to run more slowly than clocks on Earth.
- (c) It has no effect on the speed of the clocks.
- (d) It causes them to run at exactly the same speed as clocks on Earth.

問 124. In the final paragraph, what is the meaning of the word “testament”?

- (a) a written will
- (b) a serious warning
- (c) a sign or proof
- (d) a difficult challenge

問 125. What is the combined net effect of both Special and General Relativity on a GPS satellite's clock, as described in the passage?

- (a) The clocks run more slowly by about 7 microseconds per day.
- (b) The clocks run faster by about 45 microseconds per day.
- (c) The clocks run faster by a total of about 38 microseconds per day.
- (d) The clocks run at the same rate as clocks on the ground.

解答と解説

問 122

正解

(b) GPS would be useless without accounting for the effects of Einstein's theories of relativity. subsection*解説文章全体を通して、GPS が機能するためには、アインシュタインの特殊相対性理論と一般相対性理論の両方の影響を**考慮に入れ、修正する**ことが不可欠であると説明しています。

重要単語の解説:

navigate (道案内をする): 道順を決める、ナビゲーションする。

broadcast (放送する、発信する): 情報を広く送ること。

useless (役に立たない、無駄な)

問 123

subsection*正解 (b) It causes them to run more slowly than clocks on Earth. subsection*解説第 2 段落に、「特殊相対性理論 (Special Theory of Relativity) は、物体の移動速度が速いほど時間が遅く進むと述べている」とあり、「GPS 衛星は**時間のわずかな減速** (a slight slowing of time) を経験する」と説明されています。

重要単語の解説:

time dilation (時間の遅れ): 時間の進み方が遅くなる現象。

relative to (～と比較して、～に対して)

minuscule (ごくわずかな、微小な)

問 124

subsection*正解 (c) a sign or proof subsection*解説**testament** (名詞) は、ここでは「～の証 (あかし)、証拠」という意味で使われています。文章の最後では、「GPS が驚くほどの正確さで位置を特定できることは、理論物理学の実用的な力の**直接的な証拠**である」と述べられています。vspace1em

重要単語の解説:

trivial (取るに足らない、ささいな)

pinpoint (正確に特定する)

testament (証、証拠)

問 125

subsection*正解 (c) The clocks run faster by a total of about 38 microseconds per day. subsection*解説最終段落に、「修正なしでは、衛星の時計は地上にある時計より**合計で 1 日あたり約 38 マイクロ秒速く進む**」と書かれています。これは、特殊相対性理論 ($-7 \mu s$) と一般相対性理論

(+45 μ s) の効果を合わせたものです。vspace1em

重要単語の解説:

net effect (正味の効果、合計の効果)

work against each other (互いに打ち消し合う)

account for (～を考慮に入れる、～を説明する)

0.32 東京タワーの歴史

次の英文を読み、後の設問に答えなさい。

Tokyo Tower: A Symbol of Japan's Post-War Recovery

When one thinks of Tokyo, the image of Tokyo Tower often comes to mind. Standing 333 meters tall, it was once the tallest freestanding tower in the world, a title it held for over a decade. Completed in 1958, this iconic structure was built with two primary goals: to serve as a broadcast antenna for the burgeoning television industry and to stand as a powerful symbol of Japan's economic prosperity and technological prowess following the devastation of World War II.

Inspired by the Eiffel Tower in Paris, Tokyo Tower was constructed with a unique lattice design that made it lighter and more resilient than many of its contemporaries. It was built in a remarkably short period of time, thanks to the dedication and hard work of hundreds of skilled laborers. Its vibrant orange and white paint, mandated by aviation safety regulations, made it a distinctive landmark visible for miles around.

Beyond its function as a broadcasting tower, Tokyo Tower quickly became a popular tourist destination. Its two observation decks offered unparalleled panoramic views of the city, attracting millions of visitors each year. For many Japanese people, it represented a source of national pride and a hopeful look toward the future. It was a tangible sign that the nation had overcome its challenges and was ready to compete on the world stage.

However, with the construction of the Tokyo Skytree in 2012, Tokyo Tower lost its title as the city's tallest structure and primary broadcasting hub. While its practical function has diminished, its cultural significance has not. It remains an enduring symbol of a bygone era—a nostalgic reminder of a time of rapid growth and optimism. Today, it continues to draw visitors who are as much interested in its historical context as they are in its scenic views. The orange and white spire, lit up against the night sky, stands not only as a landmark but as a monument to Japan's post-war identity.

設問

- 問 126. What were the two main purposes for building Tokyo Tower, as mentioned in the first paragraph?
- (a) To be a tourist spot and a location for a restaurant.
 - (b) To be a broadcasting antenna and a symbol of national recovery.
 - (c) To be taller than the Eiffel Tower and a hub for international business.
 - (d) To be a military observation tower and a place for local events.

問 127. According to the passage, why was Tokyo Tower painted orange and white?

- (a) It was the most popular color scheme in the 1950s.
- (b) The colors were chosen to represent the Japanese flag.
- (c) It was a regulation for aviation safety.
- (d) The paint was the cheapest available at the time.

問 128. In the final paragraph, what is the meaning of the word **“bygone”**?

- (a) current and modern
- (b) past and old
- (c) difficult and challenging
- (d) hopeful and optimistic

問 129. What is the author’s overall conclusion about Tokyo Tower’s significance?

- (a) It is no longer important since the construction of the Tokyo Skytree.
- (b) Its cultural importance has increased, while its practical function has decreased.
- (c) It is still the most important broadcasting hub in Japan.
- (d) Its main purpose has always been to be a tourist attraction.

解答と解説

問 126

正解

(b) To be a broadcasting antenna and a symbol of national recovery. subsubsection*解説第 1 段落に、「テレビ放送用のアンテナ (a broadcast antenna) として機能すること」と、「日本の経済的繁栄と技術力 (a powerful symbol of Japan's economic prosperity and technological prowess) の象徴として立つこと」という 2 つの主要な目的が明記されています。

重要単語の解説:

iconic (象徴的な): ある時代や文化を代表する。

burgeoning (急成長している): 急速に発展し始めている。

prowess (卓越した能力、手腕)

問 127

subsubsection*正解 **(c) It was a regulation for aviation safety.** subsubsection*解説第 2 段落に、「航空安全規制によって義務付けられた (mandated by aviation safety regulations) 鮮やかなオレンジと白の塗装」と書かれています。vspace1em

重要単語の解説:

mandated by (～によって義務付けられた)

aviation (航空、飛行)

regulation (規制、規則)

問 128

subsubsection*正解 **(b) past and old** subsubsection*解説**bygone** (形容詞) は、「過ぎ去った、昔の」という意味です。ここでは、「過ぎ去った時代 (a bygone era)」という文脈で使われており、高度経済成長期のノスタルジックな思い出を指しています。vspace1em

重要単語の解説:

enduring (不朽の、永続的な)

nostalgic (懐かしい、ノスタルジックな)

monument (記念碑)

問 129

subsubsection*正解 **(b) Its cultural importance has increased, while its practical function has decreased.** subsubsection*解説最終段落に、「実用的な機能は衰えた (its practical function has diminished) が、文化的な重要性は失われていない (its cultural significance has not)」と明確に述べられています。vspace1em

重要単語の解説:

diminished (減少した、衰えた)

cultural significance (文化的意義)

scenic views (景色の良い眺め)

0.33 ハーバー・ボッシュ法

次の英文を読み、後の設問に答えなさい。

The Invention That Feeds the World: The Haber-Bosch Process

When we look at the world's population today, it's difficult to imagine a time when food scarcity was a constant threat. Yet, at the beginning of the 20th century, scientists were in a race to find a way to create fertilizer that could increase crop yields to feed a growing global population. The discovery that would solve this problem and change the course of human history was the **Haber-Bosch process**, a chemical reaction that allowed for the large-scale production of ammonia from atmospheric nitrogen.

Prior to this invention, farmers relied on natural sources of nitrogen, such as animal manure and mineral deposits like Chile saltpeter. However, these resources were finite and could not keep up with the demands of a rapidly industrializing world. Fritz Haber, a German chemist, successfully synthesized ammonia in a laboratory setting in 1909 by combining nitrogen gas (N_2) and hydrogen gas (H_2) under high pressure and temperature. Carl Bosch, an engineer at the chemical company BASF, later scaled up this process for industrial production.

The impact was revolutionary. Ammonia (NH_3) is a key component of synthetic fertilizers, which are essential for modern agriculture. The widespread use of these fertilizers led to a dramatic increase in agricultural productivity, famously dubbed the "Green Revolution." It is estimated that nearly half of the world's population alive today is supported by food grown using fertilizer produced by the Haber-Bosch process. This invention is credited with preventing widespread famine and enabling the population boom of the 20th century.

Despite its enormous benefits, the Haber-Bosch process is not without its environmental and social costs. The process requires vast amounts of energy, mainly from fossil fuels, making it a significant contributor to carbon emissions. Moreover, the overuse of nitrogen fertilizers can lead to soil and water pollution, causing harm to local ecosystems. Therefore, while it is a scientific triumph that has saved billions of lives, it also presents a modern challenge to find more sustainable and environmentally friendly ways to feed the world.

設問

問 130. What was the primary problem that the Haber-Bosch process was developed to solve?

- (a) The scarcity of fossil fuels for industrial production.
- (b) The inability to produce hydrogen gas for chemical reactions.
- (c) The lack of effective fertilizers to feed a growing population.

- (d) The need to find a way to store ammonia safely.

問 131. Describe the chemical reaction of the Haber-Bosch process by writing the chemical equation.

問 132. According to the passage, what is one of the negative consequences of the widespread use of the Haber-Bosch process?

- (a) It has made the world too reliant on natural resources.
- (b) It requires a lot of energy and can cause pollution.
- (c) It has led to a decrease in global food production.
- (d) It has replaced the need for farmers to grow crops.

問 133. In the final paragraph, what is the meaning of the word **triumph**?

- (a) a major failure
- (b) a great success
- (c) a small step
- (d) a difficult challenge

解答と解説

問 130

subsection*正解 **(c) The lack of effective fertilizers to feed a growing population.** subsection*解説第 1 段落に、「増え続ける地球の人口を養うために、作物の収穫量を増やすことができる肥料を作る方法を化学者たちは探していた」とあり、これがハーバー・ボッシュ法開発の主要な動機であったことが説明されています。

重要単語の解説:

scarcity (不足、欠乏): 何かが足りない状態。

fertilizer (肥料): 土壌の養分を補う物質。

crop yields (作物の収穫量)

問 131

subsection*正解 **$N_2 + 3H_2 \rightarrow 2NH_3$** subsection*解説第 2 段落に、「窒素ガス (N_2) と水素ガス (H_2) を組み合わせてアンモニア (NH_3) を合成した」とあります。化学反応式では、窒素分子 1 つと水素分子 3 つが反応して、アンモニア分子 2 つを生成します。

化学式の解説:

N_2 : 窒素分子

H_2 : 水素分子

NH_3 : アンモニア分子

問 132

subsection*正解 **(b) It requires a lot of energy and can cause pollution.** subsection*解説最終段落に、「このプロセスは、主に化石燃料から莫大な量のエネルギーを必要とし、**炭素排出の大きな要因** (significant contributor to carbon emissions) となっている」と書かれています。また、「窒素肥料の過剰な使用は、土壌や水の**汚染** (pollution) につながる」とも述べられています。vspace1em

重要単語の解説:

consequence (結果、結末): 特に、好ましくない結果を指す。

fossil fuels (化石燃料): 石炭、石油、天然ガスなど。

ecosystems (生態系): ある地域の生物と環境の相互作用システム。

問 133

subsection*正解 **(b) a great success** subsection*解説**triumph** (名詞) は「大勝利、大成功」という意味です。文脈は「何十億もの命を救った科学的な**大成功**」とポジティブにこの発明を評価しています。vspace1em

重要単語の解説:

famine (飢饉)：食料不足による大規模な飢え。

populatin boom (人口急増)

triumph (大成功、勝利)

0.34 共生、片利共生、寄生

次の英文を読み、後の設問に答えなさい。

The Intricate Dance of Life: Symbiosis in Nature

In nature, no organism exists in isolation. Species constantly interact with one another, forming complex relationships that are essential for the survival of ecosystems. These long-term interactions between different biological species are broadly known as **symbiosis**. However, not all symbiotic relationships are created equal; they can be categorized into three main types based on the effect on the organisms involved.

The first type is **mutualism**, a relationship in which both species benefit. A classic example is the relationship between bees and flowers. The bee gets nectar, a source of food, while the flower is pollinated, allowing it to reproduce. Another example is the fungi that grow on the roots of trees. The fungus helps the tree absorb water and nutrients from the soil, and in return, the tree provides the fungus with sugars produced through photosynthesis.

The second type is **commensalism**, where one species benefits while the other is neither harmed nor helped. Remora fish, for instance, have a special fin that allows them to attach to sharks. The remora benefits by gaining a free ride and scraps of food from the shark's meals, but the shark is generally unaffected. Similarly, barnacles that attach to whales get a place to live and access to nutrient-rich water without having any noticeable impact on the whale.

The final type is **parasitism**, a relationship in which one organism, the parasite, benefits at the expense of the other, the host. Ticks that feed on the blood of a dog are a clear example. The tick benefits from the meal, while the dog, the host, is harmed. Another example is a tapeworm living inside the intestines of an animal, which absorbs nutrients from the host's food, causing the host to become malnourished. Understanding these varied forms of symbiosis is key to appreciating the intricate balance of life on Earth.

設問

問 134. What is the main purpose of the passage?

- (a) To argue that all species should live in mutualistic relationships.
- (b) To describe the different types of symbiotic relationships and provide examples.
- (c) To explain how to prevent parasitic relationships in nature.
- (d) To analyze the specific diets of bees, sharks, and ticks.

問 135. According to the passage, what distinguishes a mutualistic relationship?

- (a) Only one species benefits from the relationship.
- (b) The relationship is always harmful to both species.
- (c) The relationship is short-term and accidental.
- (d) Both species involved receive a benefit.

問 136. In the second paragraph, what is an example of what the fungus receives from the tree in a mutualistic relationship?

- (a) scraps of food from the tree's meals
- (b) a free ride to a new location
- (c) sugars produced through photosynthesis
- (d) protection from insects

問 137. In the final paragraph, what is the meaning of the phrase “at the expense of”?

- (a) in addition to
- (b) without the help of
- (c) by causing harm to
- (d) for the benefit of

解答と解説

問 134

subsubsection*正解 **(b) To describe the different types of symbiotic relationships and provide examples.** subsubsection*解説本文は、共生（symbiosis）を3つの主要なタイプ（mutualism, commensalism, parasitism）に分類し、それぞれについて具体的な例を挙げて説明しています。したがって、文章の目的は共生関係の種類とその例を記述することです。

重要単語の解説:

symbiosis (共生) : 異なる生物種が長期的に相互作用する関係。

categorized (分類された) : 特定の基準に基づいてグループ分けされた。

organism (生物、有機体)

問 135

subsubsection*正解 **(d) Both species involved receive a benefit.** subsubsection*解説第2段落に、「**mutualism** (相利共生) は、両方の種が利益を得る関係である (a relationship in which both species benefit)」と明記されています。

重要単語の解説:

mutualism (相利共生) : 両方の種が利益を得る共生関係。

benefit (利益、恩恵)

pollinated (受粉した) : 花粉が運ばれ、受精を可能にすること。

問 136

subsubsection*正解 **(c) sugars produced through photosynthesis** subsubsection*解説第2段落に、「菌類は木が水と養分を吸収するのを助け、その見返りに、木は光合成によって生産された**糖分**を菌類に提供する」と述べられています。vspace1em

重要単語の解説:

fungus (菌類)

absorb (吸収する) : 液体や物質を取り込むこと。

photosynthesis (光合成) : 植物が光エネルギーを使って食物を作るプロセス。

問 137

subsubsection*正解 **(c) by causing harm to** subsubsection*解説**at the expense of**というフレーズは「～を犠牲にして」「～に損害を与えて」という意味です。寄生（parasitism）の関係は、一方の種が**もう一方に害を与えることで**利益を得るため、(c)「～に害を与えることによって」が最も近い意味になります。vspace1em

重要単語の解説:

parasite (寄生生物)：他の生物に寄生して栄養を得る生物。

host (宿主)：寄生生物が住み着く生物。

malnourished (栄養失調の)：必要な栄養を十分に摂取できていない状態。

0.35 地震とその対策

次の英文を読み、後の設問に答えなさい。

Understanding Earthquakes: From Mechanics to Global Preparedness

Earthquakes are one of nature's most powerful and unpredictable phenomena. At their core, they are a result of tectonic plate movements. The Earth's crust is made up of several large plates that are constantly in slow motion. As these plates grind against each other, immense pressure builds up at their boundaries. When this pressure is suddenly released, it sends seismic waves through the Earth, causing the ground to shake. This moment of energy release is the earthquake itself, and the point where it begins is known as the epicenter.

Different regions of the world, with their unique geological features, have developed distinct approaches to earthquake preparedness. Japan, for example, sits on a highly active seismic zone where four major tectonic plates meet. As a result, its strategy focuses on building codes that require structures to be ****resistant**** to shaking. It also emphasizes public education, with frequent earthquake drills conducted in schools and workplaces to ensure that citizens know how to react when an earthquake occurs.

In contrast, the United States has diverse seismic risks. While the West Coast, particularly California, is highly active due to the San Andreas Fault, the central and eastern parts of the country face lower, but still present, risks. The US approach combines strict building regulations in high-risk zones with advanced early warning systems. These systems detect the initial, less destructive seismic waves and send out alerts, giving people valuable seconds to take cover before the main shaking arrives.

Chile, another country with a history of powerful earthquakes, has perhaps the most stringent building codes in the world. Its strategy is based on the principle of making buildings ****ductile****, meaning they can flex and deform without collapsing. This approach has proven highly effective in withstanding massive earthquakes, minimizing fatalities even in the face of incredibly strong shaking. Ultimately, while we cannot prevent earthquakes, continuous innovation in technology and engineering allows us to mitigate their devastating impact.

設問

問 138. What is the primary cause of an earthquake, according to the passage?

- (a) The sudden release of pressure from tectonic plate movements.
- (b) The buildup of pressure from volcanic eruptions.

- (c) The grinding of tectonic plates at their boundaries.
- (d) The immediate impact of seismic waves on the Earth's surface.

問 139. How do Japan and Chile's earthquake preparedness strategies differ from each other?

- (a) Japan focuses on building resistance, while Chile focuses on building ductility.
- (b) Japan uses early warning systems, while Chile does not.
- (c) Japan prioritizes public education, while Chile focuses on rescue operations.
- (d) Japan and Chile's strategies are almost identical.

問 140. What is the purpose of the early warning systems used in the United States?

- (a) To predict the exact time and location of an earthquake.
- (b) To send out alerts after the main shaking has occurred.
- (c) To detect initial seismic waves to give people time to take cover.
- (d) To completely prevent buildings from collapsing.

問 141. In the second paragraph, what is the closest meaning of the word ***"resistant"***?

- (a) able to withstand something without being damaged
- (b) easy to break or collapse
- (c) able to predict future events
- (d) difficult to move or change

解答と解説

問 138

subsection*正解 (a) **The sudden release of pressure from tectonic plate movements.** subsection*解説第 1 段落に、「プレートが互いにこすれ合うことで巨大な圧力が蓄積し、この圧力が突然解放されることで地震波が地球を伝わり、地面が揺れる」と述べられています。この「圧力の突然の解放」が地震の直接的な原因です。

重要単語の解説:

tectonic plates (プレート): 地球の地殻を構成する巨大な板状の岩盤。

phenomenon (現象): 科学的に説明できる自然界の出来事。

epicenter (震源地): 地震が始まった地点の地表の真上。

問 139

subsection*正解 (a) **Japan focuses on building resistance, while Chile focuses on building ductility.** subsection*解説第 2 段落に、日本の戦略が「揺れに**抵抗力がある** (resistant to shaking) 構造を要求する」と書かれています。一方、最終段落には、チリの戦略が建物を「**柔軟性がある** (ductile)」ものにすることに基づいていると説明されています。

重要単語の解説:

ductile (柔軟性のある、延性のある): 力を加えても破断せずに、変形できる性質。

stringent (厳しい、厳格な): 規則や基準などが非常に厳しいこと。

mitigate (緩和する、和らげる): 悪い影響を軽減すること。

問 140

subsection*正解 (c) **To detect initial seismic waves to give people time to take cover.** subsection*解説第 3 段落に、「これらのシステムは、破壊力が小さい初期の地震波を検知し、警報を発する」と書かれています。この警報によって「主要な揺れが来る前に、人々が避難する貴重な数秒間」が与えられます。

重要単語の解説:

early warning systems (早期警報システム): 災害が起こる前に警告を発するシステム。

detect (検知する): 何かの存在を感知すること。

take cover (身を隠す、避難する): 危険から身を守るために隠れること。

問 141

subsection*正解 (a) **able to withstand something without being damaged** subsection*解説**resistant**は「抵抗力のある、耐性のある」という意味です。文脈では建物の耐震性について言及しており、「揺れに耐え、ダメージを受けずにいられる」という意味合いが最も近いです。vspace1em

重要単語の解説:

withstand (耐える、持ちこたえる): 力や圧力に屈せず、耐えること。

seismic zone (地震帯): 地震活動が活発な地域。

長文読解問題 1：ソーシャルメディアと心の健康

- **curated**: 注意深く選ばれ、構成された。
- **origin**: 起源、発生地。物事がどこから始まるかを表す名詞です。
- **phenomenon**: 現象。特に、自然界や社会で観察される出来事を指す単語です。
- **chaos**: 混乱、無秩序。
- **congestion**: 混雑、渋滞。特に交通や人混みの密集を指します。
- **retailer**: 小売業者。
- **in the red / in the black**: 赤字 (損失) である / 黒字 (利益) である。
- **consumerism**: 消費主義。物を大量に購入・消費することが良いことだとする考え方。
- **overconsumption**: 過剰消費。
- **consequence**: 結果、結末。特に、望ましくない結果を指すことが多いです。

長文読解問題 2：人工知能と創造性

- **potential**: 潜在的な可能性、潜在能力。

長文読解問題 3：ファストファッションの問題点

- **sustainable**: 環境を破壊することなく、将来にわたって維持できる。

長文読解問題 4：昆虫の擬態

- **camouflage**: 保護色。
- **entomology**: 昆虫学。

長文読解問題 5：芸術の定義と役割

- **essence**: 本質、最も重要な要素。

長文読解問題 6：大統領選

- **contentious**: 議論の的となる、争いを引き起こす。
- **unfolded**: 展開された、明らかになった。
- **breach**: 違反、侵害。
- **decorum**: 礼儀正しさ、作法。
- **manifest**: 明らかにする、表に出す。
- **rivalry**: 対立、競争。敵意を持って競い合うこと。

長文読解問題 7：ハリケーンと台風

- **tropical cyclone**: 熱帯低気圧。
- **condition**: 条件。何かが起こるために満たされるべき要件を意味します。
- **strengthen**: 強化する、勢いを増す。

長文読解問題 8：ブラックフライデーの起源と影響

- **unintended consequence**: 意図せぬ結果。予測していなかった悪い結果。
- **omake-gai**: おまけ買い。
- **significant**: 重要な、かなりの。
- **mindset**: 考え方、物の見方。
- **ethical implications**: 倫理的影響。倫理的な観点から生じる結果。
- **confront**: 直面する、立ち向かう。

長文読解問題 9：宇宙開発競争

- **supremacy**: 至上権、優位性。
- **ideological**: 思想的な、イデオロギーの。
- **ambitious**: 野心的な、意欲的な。
- **lunar**: 月の。

長文読解問題 10：小説『宇宙戦争』と社会的影響

- **widespread**: 広範囲に及ぶ、広く行き渡った。
- **susceptibility**: 影響を受けやすさ、感受性。
- **bulletin**: 速報、報道。
- **disclaimer**: 免責事項、非保証声明。ここでは「フィクションであること」を告げる部分。
- **anxiety**: 不安、心配。
- **mass hysteria**: 集団ヒステリー、集団パニック。
- **regulation**: 規制、規則。
- **journalistic**: ジャーナリズムの、報道の。
- **literacy**: 読み書き能力。ここでは、メディア情報を適切に理解・判断する能力を指す「メディアリテラシー」の意味で使われています。

長文読解問題 11：文化相対主義と自民族中心主義

- **ethnocentrism**: 自民族中心主義。自分の文化が最も優れていると見なす考え方。
- **cultural relativism**: 文化相対主義。文化には優劣がなく、それぞれを固有の文脈で理解すべきだという考え方。
- **justify**: 正当化する。
- **violate**: 違反する、侵害する。

- **universal human rights**: 普遍的人権。
- **uphold**: 堅持する、守る。
- **fundamental**: 根本的な、基本的な。

長文読解問題 12：ナスカの地上絵

- **arid**: 乾燥した、不毛な。
- **geoglyph**: 地上絵。地面に描かれた巨大な絵や模様。
- **etch**: 刻み込む、エッチングする。
- **erosion**: 浸食、腐食。
- **delicate**: 繊細な、もろい。
- **preservation**: 保存、保全。
- **monitor**: 監視する。
- **regulate**: 規制する。

長文読解問題 13：インカ帝国

- **flourish**: 繁栄する、栄える。
- **pre-Columbian**: コロンブス以前の。
- **feat**: 偉業、功績。
- **majestic**: 雄大な、荘厳な。
- **collective**: 集団的な、共同の。
- **contribute**: 貢献する、拠出する。
- **territory**: 領土、領域。
- **conquistador**: コンキスタドル（スペインの征服者）。
- **outnumbered**: 数で勝る、数が上回る。
- **firearm**: 火器、銃。
- **smallpox**: 天然痘。
- **immunity**: 免疫。
- **legacy**: 遺産、遺物。
- **indigenous**: 土着の、先住の。
- **ingenuity**: 創意工夫、巧妙さ。
- **devastating**: 壊滅的な、甚大な。

長文読解問題 14：大航海時代と対数の歴史

- **navigators**: 航海士、ナビゲーター。
- **astronomers**: 天文学者。
- **time-consuming**: 時間のかかる、手間のかかる。
- **incentive**: 動機、誘因。
- **prone to**: ～しがちである、～の傾向がある。
- **transform**: 変形させる、変換する。
- **genius**: 天才性、非凡な才能。

- **look up**: (辞書などで) 調べる。
- **innovation**: 革新、イノベーション。
- **profound**: 深遠な、重大な。
- **progress**: 進歩、発展。

長文読解問題 15：超伝導とリニアモーターカー

- **resistance**: 抵抗。
- **expel**: 追い出す、排斥する。
- **levitate**: 浮揚する、宙に浮く。
- **conventional**: 従来の、一般的な。
- **suspend**: 吊り下げる、宙に浮かせる。
- **coil**: コイル。
- **propel**: 推進する、前進させる。
- **widespread**: 広範囲にわたる、普及した。
- **infrastructure**: インフラ、社会基盤。
- **accessible**: 利用しやすい、入手しやすい。

長文読解問題 16：日米の高校生活の対比

- **underlying**: 根底にある、根本的な。
- **philosophy**: 哲学、考え方。
- **glimpse**: ちらりと見ること。
- **academic**: 学問的な、学業の。
- **curriculum**: カリキュラム、教育課程。
- **standardized**: 標準化された。
- **rigorous**: 厳格な、厳しい。
- **specialize**: 専門とする、特化する。
- **elective**: 選択科目。
- **harmony**: 調和。
- **conformity**: 協調性、順応性。
- **individualism**: 個人主義。
- **self-expression**: 自己表現。

長文読解問題 17：大学の教育システム

- **entail**: ～を伴う、～を必要とする。
- **informed**: 情報に基づいた、十分な知識を持った。
- **holistic**: 全体的な、包括的な。
- **discipline**: 学問分野。
- **transferable skills**: 応用可能なスキル。
- **critics**: 批判者。
- **geared towards**: ～向けに作られた。

- **collaborate**: 協力する、協働する。

長文読解問題 18：ローマ帝国の衰退

- **dominant**: 支配的な。
- **decline**: 衰退、衰え。
- **complex**: 複雑な。
- **combination**: 組み合わせ。
- **internal**: 内部の。
- **external**: 外部の。
- **demise**: 終焉、崩壊。
- **instability**: 不安定さ。
- **central authority**: 中央権力。
- **bureaucracy**: 官僚制度。
- **barbarian**: 異民族の、未開の。
- **frontier**: 国境、辺境。

長文読解問題 19：日本の高度経済成長

- **unprecedented**: 前例のない。
- **transform**: ～を変形させる、～を一変させる。
- **numerous**: 多数の。
- **recovery**: 回復、立ち直り。
- **driver**: 推進力、原動力。
- **collaboration**: 協力、協業。
- **competitive**: 競争力のある。
- **industrialization**: 工業化。
- **significant**: 重大な、重要な。
- **urban**: 都市の。
- **inadequate**: 不十分な、貧弱な。

長文読解問題 20：日本の四大公害病

- **legacy**: 遺産。過去から受け継がれたもの。良いものにも悪いものにも使われる。
- **unregulated**: 規制されていない。法律や規則による制限がないこと。
- **devastating**: 破壊的な、壊滅的な。非常に大きな損害を与えるさま。
- **plight**: 苦境、窮状。
- **contaminated**: 汚染された。有害な物質などが混入していること。
- **accumulate**: 蓄積する。徐々に集まっていくこと。
- **neurological disorders**: 神経障害。脳、脊髄、神経に影響を及ぼす病気。
- **wake-up call**: 警鐘。問題に気づかせ、行動を促すきっかけ。
- **stricter**: より厳しい。
- **compensate**: 補償する。損害や苦痛に対して金銭などを支払うこと。

長文読解問題 21：AI と倫理

- **integrated**: 統合された。複数のものが一つにまとめられている状態。
- **dilemmas**: ジレンマ。二つの選択肢があり、どちらを選んでも望ましくない結果になる板挟みの状況。
- **ethical**: 倫理的な。道徳や規範に関わるさま。
- **scenario**: シナリオ。想定される状況や筋書き。
- **utilitarianism**: 功利主義。最大多数の最大幸福を目指す倫理思想。
- **framework**: 枠組み。思考や行動の土台となる構造。
- **endanger**: 危険にさらす。
- **explicit**: 明白な、明確な。
- **instinctively**: 本能的に。
- **pre-programmed**: 事前にプログラムされた。

長文読解問題 22：流行と集団心理

- **trend**: 流行、傾向。一時的に広まる社会現象。
- **rooted**: 根付いた。深く結びついていること。
- **collective behavior**: 集団行動。多くの人が似た行動をとること。
- **innate**: 生来の、生まれつきの。もともと備わっている性質。
- **conformity**: 同調。周囲に合わせて行動すること。
- **outcast**: のけ者、仲間外れ。集団から外された人。
- **adhere**: 固執する、忠実に守る。
- **paradoxical**: 逆説的な。一見矛盾しているように見えるさま。
- **individuality**: 個性、独自性。その人や物だけが持つ特徴。
- **subculture**: サブカルチャー。主流文化とは異なる、特定の集団が共有する文化。
- **authentic**: 本物の、本当の。偽りや見せかけではないさま。

長文読解問題 23：日本のアニメの海外影響

- **vehicle**: 手段、媒体。
- **niche interest**: ニッチな興味。特定の少数の人々にしか知られていない興味。
- **prominence**: 卓越、著名。広く知られていること、目立つこと。
- **globalization**: グローバル化。世界が一体となること。
- **foster**: 育む、促進する。
- **diverse backgrounds**: 多様な背景。人種、文化、国籍などが異なること。
- **bridge**: 橋渡しをする。隔たりを埋める。
- **distinctive**: 独特の、特徴的な。
- **aesthetic**: 美的、美学。美の観点。
- **merchandise**: 商品、グッズ。

長文読解問題 24：おまけと社会問題

- **phenomenon**: 現象。注目すべき出来事や事象。
- **consumerism**: 消費主義。物質的な消費を重んじる考え方。
- **unintended consequence**: 意図せぬ結果。予測していなかった悪い結果。
- **primary goal**: 第一の目的。
- **miniature**: ミニチュア。小さな模型や人形。
- **discarded**: 捨てられた。
- **significant**: 重要な、かなりの。
- **mindset**: 考え方、物の見方。
- **ethical implications**: 倫理的影響。倫理的な観点から生じる結果。
- **confront**: 直面する、立ち向かう。

長文読解問題 25：クリントンとブッシュの移行期

- **transition**: 移行。ある政権から別の政権へ移ること。
- **prank**: いたずら、悪ふざけ。
- **rivalry**: 対立、競争。敵意を持って競い合うこと。
- **intentional mischief**: 意図的ないたずら。
- **vandalism**: 器物損壊。公共物や他人の物を意図的に壊す行為。
- **obscene**: わいせつな、不快な。
- **scrawled**: 乱雑に書かれた。
- **contentious**: 議論の的となる、争いを引き起こす。
- **unfolded**: 展開された、明らかになった。
- **breach**: 違反、侵害。
- **decorum**: 礼儀正しさ、作法。
- **manifest**: 明らかにする、表に出す。

長文読解問題 26：世界の原発事故

- **distinct**: 明確に異なる、別個の。
- **profoundly**: 深く、大いに。影響や変化の度合いが非常に大きいこと。
- **perception**: 認識、見方。
- **crucially**: 決定的に、極めて重要に。
- **containment building**: 格納容器。原子炉を覆い、放射性物質の漏洩を防ぐための建物。
- **vulnerability**: 脆弱性、弱点。
- **overwhelm**: 圧倒する、制圧する。
- **preparedness**: 準備、備え。
- **grim**: 厳しい、暗い。希望がなく、悲惨な様子。
- **transparency**: 透明性。情報が隠されずに公開されていること。
- **disregarded**: 無視した、軽視した。

長文読解問題 27：ベトナム戦争とペンタゴン・ペーパーズ

- **deception**: 欺瞞、詐欺。人を欺く行為。
- **lasting**: 永続的な。長く続く、恒久的な。
- **fundamentally**: 根本的に。根底から、抜本的に。
- **commissioned**: 依頼された、委託された。
- **disillusioned with**: ～に幻滅した。
- **escalating**: エスカレートさせていた、拡大させていた。
- **divisive**: 意見の対立を引き起こす、分裂を招く。
- **turmoil**: 騒乱、混乱。社会的な動揺や無秩序な状態。
- **injunction**: 差し止め命令。裁判所が出す、特定の行為を禁じる命令。
- **landmark**: 画期的な。歴史に残るほど重要な出来事。
- **upholding**: 支持する、維持する。物事の正当性や原則を守ること。

長文読解問題 28：パディントンという名のクマ

- **enduring**: 永続的な。長く続く、不朽の。
- **profound**: 深遠な。表面的なものではなく、深い意味や影響を持つ。
- **commentary**: 論評、解説。特定の主題についての意見や考察。
- **displaced person**: 避難民、難民。戦争や災害などで故郷を追われた人々。
- **metaphor**: 比喩、暗喩。ある事柄を別の事柄になぞらえて表現する修辞法。
- **compassion**: 思いやり、同情。
- **tolerance**: 寛容。自分と異なる考えや行動を受け入れること。
- **empathy**: 共感。他者の感情や経験を理解し、共有すること。
- **scorn**: 軽蔑する、あざ笑う。
- **embodied**: 体現する。思想や性質などを具体的な形で表すこと。
- **dual identity**: 二重のアイデンティティ。
- **fluid**: 流動的な、固定されていない。

長文読解問題 29：彗星の謎

- **unveiling**: ～の秘密を明らかにする。隠されていたものや謎を解き明かすこと。
- **celestial**: 天体の、空の。
- **harbinger**: 先触れ、前兆。何かが起こる兆候となるもの。
- **solar radiation**: 太陽放射。太陽から放出されるエネルギー。
- **solar wind**: 太陽風。太陽から放出される荷電粒子の流れ。
- **radiation pressure**: 放射圧。光子が物体に与える圧力。
- **sublimate**: 昇華する。固体が液体を経由せず直接気体に変化すること。
- **coma**: コマ。彗星の核の周りにあるガス状の大気。
- **characteristic**: 特徴的な。
- **elliptical**: 楕円形の。天体の軌道を表す言葉。
- **spherical**: 球状の。

- **Oort Cloud**: オールトの雲。太陽系の最も外側にある、彗星の故郷とされる仮説上の領域。

長文読解問題 30：量子力学と半導体

- **abstract**: 抽象的な。具体性や実体がなく、観念的な。
- **counterintuitive**: 直感に反する。直感的に理解しにくい、意外な。**fundamental**: 基礎的な、根本的な。
- **conductor**: 導体。電気を通しやすい物質。
- **insulator**: 絶縁体。電気を通しにくい物質。
- **energy gap**: エネルギーギャップ。物質中の電子が存在できないエネルギーの範囲。
- **harnessed**: 利用された、制御された。
- **revolution**: 革命。ある分野における根本的な変化。
- **supercomputer**: スーパーコンピューター。
- **transistor**: トランジスタ。電気信号を増幅またはスイッチングする半導体素子。
- **precision**: 精密さ、正確さ。
- **off and on**: オンとオフ。

長文読解問題 31：相対性理論と GPS

- **navigate**: 道案内をする。道順を決める、ナビゲーションする。
- **broadcast**: 放送する、発信する。情報を広く送ること。
- **useless**: 役に立たない、無駄な。
- **time dilation**: 時間の遅れ。時間の進み方が遅くなる現象。
- **relative to**: ～と比較して、～に対して。
- **minuscule**: ごくわずかな、微小な。
- **trivial**: 取るに足らない、ささいな。
- **pinpoint**: 正確に特定する。
- **testament**: 証、証拠。
- **net effect**: 正味の効果、合計の効果。
- **work against each other**: 互いに打ち消し合う。
- **account for**: ～を考慮に入れる、～を説明する。

長文読解問題 32：東京タワーの歴史

- **iconic**: 象徴的な。ある時代や文化を代表する。
- **burgeoning**: 急成長している。急速に発展し始めている。
- **prowess**: 卓越した能力、手腕。
- **mandated by**: ～によって義務付けられた。
- **aviation**: 航空、飛行。
- **regulation**: 規制、規則。
- **bygone**: 過ぎ去った、昔の。
- **enduring**: 不朽の、永続的な。
- **nostalgic**: 懐かしい、ノスタルジックな。

- **monument**: 記念碑。
- **diminished**: 減少した、衰えた。
- **cultural significance**: 文化的意義。
- **scenic views**: 景色の良い眺め。

長文読解問題 33：ハーバー・ボッシュ法

- **scarcity**: 不足、欠乏。何かが足りない状態。
- **fertilizer**: 肥料。土壌の養分を補う物質。
- **crop yields**: 作物の収穫量。
- **consequence**: 結果、結末。特に、好ましくない結果を指す。
- **fossil fuels**: 化石燃料。石炭、石油、天然ガスなど。
- **ecosystems**: 生態系。ある地域の生物と環境の相互作用システム。
- **triumph**: 大成功、勝利。
- **famine**: 飢饉。食料不足による大規模な飢え。
- **population boom**: 人口急増。

長文読解問題 34：共生、片利共生、寄生

- **symbiosis**: 共生。異なる生物種が長期的に相互作用する関係。
- **categorized**: 分類された。特定の基準に基づいてグループ分けされた。
- **organism**: 生物、有機体。
- **mutualism**: 相利共生。両方の種が利益を得る共生関係。
- **benefit**: 利益、恩恵。
- **pollinated**: 受粉した。花粉が運ばれ、受精を可能にすること。
- **fungus**: 菌類。
- **absorb**: 吸収する。液体や物質を取り込むこと。
- **photosynthesis**: 光合成。植物が光エネルギーを使って食物を作るプロセス。
- **parasite**: 寄生生物。他の生物に寄生して栄養を得る生物。
- **host**: 宿主。寄生生物が住み着く生物。
- **malnourished**: 栄養失調の。必要な栄養を十分に摂取できていない状態。

長文読解問題 35：地震とその対策

- **tectonic plates**: 地球の地殻を構成する巨大な板状の岩盤。
- **phenomenon**: 科学的に説明できる自然界の出来事。
- **epicenter**: 地震が始まった地点の地表の真上。
- **ductile**: 柔軟性のある、延性のある。力を加えても破断せずに、変形できる性質。
- **stringent**: 厳しい、厳格な。規則や基準などが非常に厳しいこと。
- **mitigate**: 緩和する、和らげる。悪い影響を軽減すること。
- **early warning systems**: 早期警報システム。災害が起こる前に警告を発するシステム。
- **detect**: 検知する。何かを存在を感知すること。
- **take cover**: 身を隠す、避難する。危険から身を守るために隠れること。

- **resistant**: 抵抗力のある、耐性のある。揺れに耐え、ダメージを受けずにいられること。
- **withstand**: 耐える、持ちこたえる。力や圧力に屈せず、耐えること。 **seismic zone**: 地震帯。地震活動が活発な地域。